THEORY OF QUANTUM NOISE AND DECOHERENCE

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1. Recapitulation of quantum mechanics

1.1. Why quantum noise and decoherence? You have all met quantum mechanics before—it is a prerequisite for this course. But you have probably only encountered the quantum mechanics of *closed* systems: free particles, harmonic oscillators, and other idealised setups that do not interact with their environment. The objective of this course is to go beyond this and study the quantum mechanics of *open* systems—quantum systems composed of many subsystems.

In contrast to classical physics, quantum information *cannot be copied*: the **no-cloning theorem** prevents us from duplicating quantum states. If we want to gain information about a closed quantum system, we have to interact with it, and this interaction unavoidably has some back-action on the system itself. *The act of measuring disturbs.* This is the defining feature of quantum mechanics.

Measurement disturbance in everyday life

Measurement back-action is not unique to quantum mechanics. Two classical examples:

- **Stock markets:** acting on information about a stock creates disturbance, since other traders observe and react to your trades.
- **Observer effect in social psychology:** subjects who know they are being observed behave differently.

In classical physics, these are consequences of approximations. In quantum mechanics, measurement disturbance is *fundamental*—it is built into the theory itself.

By the end of this course, we aim to understand this statement quantitatively, and also to explain *why quantum systems look classical* when we observe them with our everyday measurement apparatus. The answer we propose: because of **quantum noise and decoherence**, which are themselves a consequence of the fact that measuring disturbs a quantum system.

1.2. Course overview. The course covers the following topics:

(1) Recapitulation of quantum mechanics (complementarity and uncertainty, quantitative statements, entanglement, entropy); 1–2 lectures.
(2) Complete positivity and dilation theory (quantum channels, POVM observables, state after general measurements, non-demolition measurements, Naimark, Stinespring); 1 lecture.
(3) Quantum many-particle systems (Fock space, quasi-free states, mean-field limits); 2 lectures.
(4) Continuous variables (covariance matrices, Gaussian states, Wigner distributions, examples from quantum optics, standard quantum limit, continuous variables from collective degrees of freedom); 2 lectures.
(5) Lindblad equations (physical background: quantum dynamics with dissipation and decoherence, Markov approximation, derivations: weak and strong coupling regime); 2 lectures.
(6) Decoherence sources in experiments (the basic mechanism: losing parts of the system, decoherence of Schrödinger cat states, classical limit); 1 lecture.

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(7) Classical noise and stochastic calculus (basics of stochastic processes, counting processes, photodetection, factorial moments, Markov chains and processes, martingales, Langevin and Fokker–Planck equations); 2 lectures.
(8) Quantum stochastic calculus (discrete case: matrix product states, unravelling of Lindblad equations, quantum statistics of the output events, continuous quantum measurement, cavity input–output formalism).
(9) Feedback and control (classical and quantum feedback, Lie algebra techniques, controllability, linear quantum control systems); 1 lecture.

Books/Notes:

(1) Reinhard — Introduction to QM + QI [Wer12]
(2) B. Davies [Dav76]
(3) M. Wolf's notes [Wol12]
(4) Gardiner and Zoller [GZ04]
(5) Gardiner [Gar09]
(6) Wiseman & Milburn [WM10]
(7) Walls & Milburn [WM08]

1.3. Kinematics of quantum mechanics.

We divide physical experiments into two parts:

In classical physics there is no real need to talk of measurements. In statistical theories, such as quantum mechanics, all we can do is predict probabilities of outcomes in statistical experiments.

Definition 1.1 (Preparation)**.** A *preparation* of a quantum system is the set of actions which determines all probability distributions of any possible measurement. Different preparations can lead to identical probability distributions $\Rightarrow$ we introduce the concept of *state*, which specifies only the effect of preparation regardless of how it was performed.

State

A *state* describes all details relevant for subsequent measurement.

Mathematically, the division of experiments into preparations and measurements of observables is represented, in quantum mechanics, as follows.

1.3.1. *Observables.*

Observables are hermitian elements of an algebra $\mathcal{A}$ (think: matrices of side n).

An *algebra* is a set closed under multiplication and addition as well as multiplication by scalars. Each element $A \in \mathcal{A}$ has an adjoint $A^\dagger \in \mathcal{A}$. An element $E \in \mathcal{A}$ is *positive* if $E = A^\dagger A$.

The algebra $\mathcal{A}$ is usually represented in terms of bounded linear operators $\mathcal{B}(\mathcal{H})$ acting on a Hilbert space $\mathcal{H}$.

Example 1.1 (Qubit observables)**.** The set of observables for a *qubit* are hermitian 2×2 matrices in

$$\mathcal{A}_n \cong \mathcal{A} \cong M_2(\mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \,\middle|\, a,b,c,d \in \mathbb{C} \right\}, \qquad \mathcal{A}_h = \{X \in \mathcal{A} \mid X^\dagger = X\}.$$

1.3.2. *Effects and measurements.* Measurements themselves (not observables) are described by assigning to each *outcome* of a device an *effect* $E \in \mathcal{A}$: an element satisfying

$$0 \leq E \leq \mathbb{I}\,. \tag{1}$$

Here $A \leq B$ means $B - A = X^\dagger X$ for some X.

Example 1.2 (Photon counter)**.** A photon counter has two possible outcomes: no click or click. The corresponding effects are:

$$E_{\text{no click}} = |0\rangle\langle 0|\,, \qquad E_{\text{click}} = \mathbb{I} - |0\rangle\langle 0|\,, \tag{2}$$

where $|0\rangle$ is the vacuum state. Homodyne detection, by contrast, assigns to each outcome a projection onto a coherent state—an example of a POVM that does *not* arise from a standard textbook observable.

1.3.3. *States.* A *state* ω on $\mathcal{A}$ is a positive normalised linear functional on $\mathcal{A}$. That is:

$$\omega\colon \mathcal{A} \to \mathbb{C} \text{ is linear}, \quad \omega(X^\dagger X) \geq 0, \quad \omega(\mathbb{I}) = 1\,. \tag{3}$$

We think of states as assigning probabilities to outcomes of measurements. Each of the three axioms has a physical origin:

- **Normalisation** $\omega(\mathbb{I}) = 1$: the identity is the effect corresponding to "doing nothing whatsoever." Its probability must be 1 (probability is conserved).
- **Positivity** $\omega(X^\dagger X) \geq 0$: if we put in a valid measurement (a positive operator), we must get out a valid probability (a non-negative number).
- **Linearity**: if we randomise between two preparations (e.g. flip a fair coin and perform preparation A on heads, preparation B on tails), the resulting state is a convex combination $\frac{1}{2}\omega_A + \frac{1}{2}\omega_B$. Linearity ensures this is consistent.

Remark. States described as rays in Hilbert space are a *subset* of all possible states: they are the pure states. In this course we deal with the real world, where noise and imperfections force quantum systems into *mixed* states, and we therefore need the full generality of the algebraic framework.

1.3.4. *Probabilities and the Born rule.* The probability of outcome E measured on a state ω is

$$\boxed{p(E) = \omega(E)}\,. \tag{4}$$

Label outcomes corresponding to observable A by index α. We assign an effect $0 \leq E_\alpha \leq \mathbb{I}$ to each α so that $p_\alpha =$ prob. of outcome α can be calculated via

$$p_\alpha = \omega(E_\alpha)\,. \tag{5}$$

In order that $\sum_\alpha p_\alpha = 1$, we require

$$\boxed{\sum_\alpha E_\alpha = \mathbb{I}}\,. \tag{6}$$

Definition 1.2 (POVM)**.** A collection $\{E_\alpha\}$ of effects such that $\sum_\alpha E_\alpha = \mathbb{I}$ is called a *POVM* (positive operator-valued measure).

1.3.5. *Connection to textbook observables.* The connection between textbook observables A and POVMs is via the spectral decomposition:

$$A = \sum_\alpha a_\alpha \, E_\alpha \,, \tag{7}$$

where a_α are the eigenvalues and E_α are the spectral projectors. Note $\sum_\alpha E_\alpha = \mathbb{I}$.

POVMs are *more general*: the E_α need not be orthogonal projectors. Every observable gives rise to a POVM (via its spectral decomposition), but not every POVM comes from an observable. This extra generality is very useful: measurements such as homodyne and heterodyne detection in quantum optics do *not* correspond to measuring a standard hermitian observable—they are inherently POVM measurements.

Why POVMs matter

It is somewhat puzzling that textbook quantum mechanics emphasises "observables correspond to hermitian operators," when in practice many real experiments (heterodyne detection, weak measurements, etc.) perform measurements that are strictly more general than projective measurements of a hermitian operator. The POVM framework captures *all* possible measurements in a unified way.

1.4. Density operators. For finite-dimensional systems, e.g. qubits, we can *always* represent states $\omega\colon \mathcal{A} \to \mathbb{C}$ as *density operators* via

$$\boxed{\omega(A) = \mathrm{tr}(\rho\, A)} \,, \tag{8}$$

where

$$\mathrm{tr}(\rho) = 1 \qquad \text{and} \qquad \rho \geq 0 \,. \tag{9}$$

Linearity of the state is automatic, since the trace is linear.

Remark. The identification of states with density operators only works reliably for finite-dimensional systems. In infinite dimensions (e.g. the electromagnetic field), there exist operators whose trace diverges, and the set of states is strictly larger than the set of density operators. This is why we emphasise the algebraic definition of states as functionals on observables—it remains valid in all dimensions.

Definition 1.3 (Pure state)**.** A state ω is *pure* if it cannot be written as a nontrivial convex combination (i.e. with $p \notin \{0, 1\}$):

$$\omega = (1-p)\,\omega' + p\,\omega'' \,. \tag{10}$$

For finite-dimensional systems, ω is pure if and only if

$$\omega(\cdot) = \mathrm{tr}(\rho\,\cdot) \quad \text{with} \quad \rho = |\psi\rangle\langle\psi| \,, \quad |\psi\rangle \in \mathcal{H} \,. \tag{11}$$

That is, a pure state corresponds to a rank-one projector—a "ray" in Hilbert space.

Example 1.3 (Qubit states and the Bloch sphere)**.** All qubit states ω may be represented as 2×2 matrices $\rho \geq 0$, $\mathrm{tr}(\rho) = 1$. The most general such matrix that satisfies $\mathrm{tr}(\rho) = 1$ can be parameterised as:

$$\rho = \frac{1}{2}\begin{pmatrix} 1 + x_3 & x_1 - i x_2 \\ x_1 + i x_2 & 1 - x_3 \end{pmatrix} = \frac{\mathbb{I} + \boldsymbol{x}\cdot\boldsymbol{\sigma}}{2} \,, \tag{12}$$

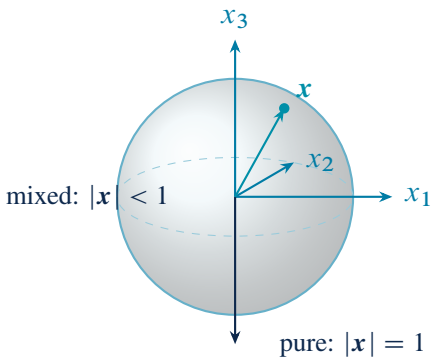
where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \boldsymbol{x} \in \mathbb{R}^3 \,. \tag{13}$$

Exercise 1.1. Show that $\rho \geq 0 \;\Rightarrow\; |\boldsymbol{x}|^2 \leq 1$.

Hint: Compute the eigenvalues of ρ (they are p and $1-p$ for some $p \in [0,1]$) and impose $\rho \geq 0$. Alternatively, use the constraint $\rho^2 = \rho$ for pure states to show that pure states satisfy $|\boldsymbol{x}| = 1$, and then argue that mixed states lie in the interior by convexity.

Thus $\boldsymbol{x}$ lies in the unit ball of radius 1 in $\mathbb{R}^3$—the *Bloch ball*—and the density operator ρ is in one-to-one correspondence with $\boldsymbol{x}$. Pure states live on the *surface* of the sphere ($|\boldsymbol{x}| = 1$); mixed states occupy the interior.



The Bloch sphere

It is a remarkable fact that qubit states—abstract 2×2 positive matrices with unit trace—admit a geometric representation as points in a three-dimensional ball. Operations on qubits (rotations, measurements, decoherence) all have clean geometrical interpretations in terms of this object. Sadly, for qutrits (3-level systems) and beyond, no comparably simple geometric picture exists: the state space becomes a complicated convex body in $d^2 - 1$ dimensions with "horrible corners and curvy bits."

Example 1.4 (Effect measurement on a qubit)**.** Consider the effect

$$E = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \tfrac{1}{2}(\mathbb{I} + \sigma_z)\,.$$

This corresponds to measuring spin-up in the z-direction. The probability of this outcome is

$$p(E) = \omega(E) = \operatorname{tr}(\rho\, E) = \tfrac{1}{2}(1 + x_3)\,.$$

Geometrically, this probability is determined by the projection of the Bloch vector $\boldsymbol{x}$ onto the x_3-axis.

1.5. Composite systems and entanglement. The observables for a composite quantum system AB with observable algebras $\mathcal{A}$ and $\mathcal{B}$, respectively, is

$$\mathcal{A} \otimes \mathcal{B}\,. \tag{14}$$

The underlying Hilbert space for AB is a tensor product as well:

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B\,. \tag{15}$$

The simplest states ω of a composite system AB are *products*, i.e.

$$\omega(X \otimes Y) = \omega_A(X) \cdot \omega_B(Y)\,, \qquad X \in \mathcal{A},\; Y \in \mathcal{B}\,. \tag{16}$$

Pure product states ω have density operator

$$\rho = |\psi\rangle\langle\psi| \otimes |\phi\rangle\langle\phi|\,, \qquad |\psi\rangle \in \mathcal{H}_A,\; |\phi\rangle \in \mathcal{H}_B\,. \tag{17}$$

States which are *not* product are *correlated*.

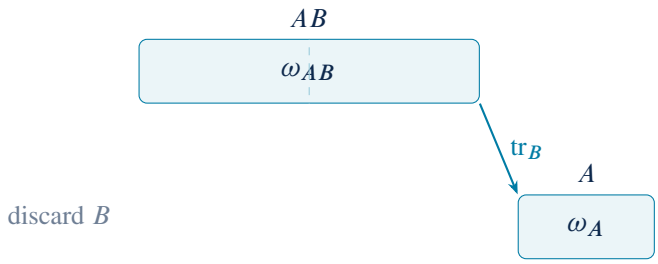
1.5.1. *Reduced states and partial trace.* The state of AB, when restricted to observables acting nontrivially only on A, i.e. observables of the form $X_A \otimes \mathbb{I}_B$, is called the *reduced state* ω_A. The corresponding *reduced density operator* ρ_A on $\mathcal{A}$ is defined by

$$\omega_A(X) = \mathrm{tr}[\rho_A X] = \mathrm{tr}[\rho\,(X \otimes \mathbb{I})] \qquad \forall\, X \in \mathcal{A}\,. \tag{18}$$

The map

$$\mathrm{tr}_B \colon \rho \mapsto \rho_A := \sum_j (\mathbb{I} \otimes \langle j|)\,\rho\,(\mathbb{I} \otimes |j\rangle) \quad , \tag{19}$$

where $\{|j\rangle\}$ is an ONB of $\mathcal{H}_B$, is called the *partial trace*; ρ_A is the *marginal* of ρ.



1.6. Schmidt decomposition.

Proposition 1.1 (Schmidt decomposition)**.** *For every vector* $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$*, with* $d = \min\{\dim(\mathcal{H}_A), \dim(\mathcal{H}_B)\}$*, there exists an ONB* $\{|g_j\rangle\} \subset \mathcal{H}_A$ *and* $\{|f_j\rangle\} \subset \mathcal{H}_B$ *such that*

$$|\psi\rangle = \sum_{j=1}^{d} \sqrt{p_j}\,|g_j\rangle \otimes |f_j\rangle \quad , \qquad p_j \geq 0, \quad \sum_j p_j = \||\psi\rangle\|^2\,. \tag{20}$$

The $\sqrt{p_j}$ *are called the* Schmidt coefficients*; the number of nonzero* p_j *is the* Schmidt rank.

Proof. Write $|\psi\rangle$ in a product basis:

$$U \otimes V\big(|\psi\rangle\big) = U \otimes V\Big(\sum_{j,k} [\Psi]_{jk}\,|j\rangle|k\rangle\Big) = \sum_{j,j'}\sum_{k,k'} [U]_{jj'}\,[\Psi]_{j'k'}\,[V]_{k'k}\,|j\rangle|k\rangle\,.$$

By the singular value decomposition (SVD), choose unitary U, V such that $U\Psi V^\top = D$ is diagonal with non-negative entries. Then

$$|\psi\rangle = \sum_j [D]_{jj}\,|g_j\rangle \otimes |f_j\rangle\,,$$

where $|g_j\rangle = U^\dagger|j\rangle$ and $|f_j\rangle = V^\dagger|j\rangle$ form orthonormal bases of $\mathcal{H}_A$ and $\mathcal{H}_B$ respectively. Setting $\sqrt{p_j} = [D]_{jj}$ gives the Schmidt form. □

1.7. Entanglement versus classical correlations. We saw in Lecture 1 that the Schmidt decomposition tells us when a *pure* state of a composite system is entangled: it is entangled if and only if its Schmidt rank is greater than one. But for *mixed* states, the situation is more subtle.

Consider the two-qubit density operator

$$\rho = p\,|00\rangle\langle 00| + (1-p)\,|11\rangle\langle 11|\,, \qquad p \in (0,1)\,. \tag{21}$$

This is *not* a product state: $\rho \neq \rho_A \otimes \rho_B$ for any pair of single-qubit states ρ_A, ρ_B. Yet it is manifestly *not entangled*—it can be prepared by a purely classical procedure:

Correlated coins

Flip a (biased) coin. If heads (probability p), prepare both qubits in $|0\rangle$; if tails (probability $1-p$), prepare both in $|1\rangle$. Hand one qubit to Alice and the other to Bob. The joint state is (21)—correlated, but only classically so.

A state is called **separable** (or *classically correlated*) if it can be written as a convex combination of product states:

$$\rho = \sum_k \lambda_k \, \rho_A^{(k)} \otimes \rho_B^{(k)}, \qquad \lambda_k \geq 0, \quad \sum_k \lambda_k = 1\,. \tag{22}$$

A state that is *not* separable is **entangled**.

Remark. For pure states, "separable" reduces to "product," and entanglement can be efficiently decided via the Schmidt decomposition. For mixed states, deciding whether a given ρ is separable or entangled is, in general, a computationally hard problem and an active area of research in quantum information theory.

1.7.1. *Entanglement and quantum noise.* There is a deep connection between entanglement and quantum noise that motivates this entire course. In classical probability, a pure state of a composite system always assigns definite (pure) states to each subsystem. In quantum mechanics, this is spectacularly *false*:

Entanglement ⇔ noise on subsystems

A composite quantum system can be in a pure state (zero temperature, unitary evolution, no noise whatsoever), yet its subsystems are in *mixed* states—they look noisy and are decohering rapidly. This happens precisely when the global state is entangled.

Recall from the Schmidt decomposition (Proposition 1.1) that for any pure state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ with Schmidt form $|\psi\rangle = \sum_j \sqrt{p_j}\,|g_j\rangle \otimes |f_j\rangle$, the reduced state of subsystem A is

$$\rho_A = \mathrm{tr}_B(|\psi\rangle\langle\psi|) = \sum_j p_j \, |g_j\rangle\langle g_j|\,. \tag{23}$$

This is pure if and only if exactly one p_j is nonzero—i.e. when $|\psi\rangle$ is a product state. Otherwise, ρ_A is mixed: the subsystem has acquired "noise" entirely from entanglement with the rest of the system.

This is, in some sense, the central theme of this course: noise comes from entanglement with an environment, and quantum computation exploits entanglement within a system. Making a quantum computer work requires fighting entanglement with entanglement.

1.8. Dynamics of quantum systems. A good physical theory has both *kinematics* (how to represent states and observables) and *dynamics* (how to represent the passage of time). The fundamental dynamical question is:

> *If I know how to represent a measurement at time $t = 0$, how do I represent the same measurement at time t?*

We divide a generic experiment into three stages:

1.8.1. *Heisenberg and Schrödinger pictures.* There are two complementary ways to absorb the evolution back into the kinematics:

- **Schrödinger picture:** the evolution produces a new effective state ω' from the old state ω; observables are unchanged.
- **Heisenberg picture:** the evolution produces a new effective observable A' from the original A; the state is unchanged.

Both pictures describe the same physics. Their compatibility is expressed by requiring that for every state ρ and observable A,

$$\operatorname{tr}\big(\rho\ \mathcal{U}(A)\big) = \operatorname{tr}\big(\mathcal{U}_*(\rho)\ A\big)\,, \tag{24}$$

where $\mathcal{U}$ is the Heisenberg-picture evolution and $\mathcal{U}_*$ is its Schrödinger-picture dual.

Remark. There are also *interaction pictures* in which part of the evolution is applied to the state and part to the observable. These are useful for perturbation theory but will not play a role in the first part of this course.

1.8.2. *Requirements for evolution maps.* What are the minimal requirements for a map to qualify as a physical evolution? Working in the Heisenberg picture, an evolution $\mathcal{U}\colon \mathcal{A} \to \mathcal{A}$ must satisfy:

(i) **Linearity:** $\mathcal{U}$ is a linear map. This follows from the requirement that randomising over preparations before or after the evolution gives the same result.
(ii) **Probability preservation (unitality):** $\mathcal{U}(\mathbb{I}) = \mathbb{I}$. If the effect "do nothing" is evolved, it must remain "do nothing"—otherwise probability is created or destroyed.
(iii) **Positivity:** if $E \geq 0$ then $\mathcal{U}(E) \geq 0$. Legal effects must be mapped to legal effects.

1.8.3. *Closed systems: unitary evolution.* A *closed* (or *reversible*) evolution is one that, in addition to the above, is invertible. For finite-dimensional systems, closed evolutions are exactly those representable in terms of a unitary operator U:

$$\boxed{\mathcal{U}(A) = U^\dagger A\, U}\,, \qquad U^\dagger U = U U^\dagger = \mathbb{I}\,. \tag{25}$$

In the Schrödinger picture, the corresponding map is

$$\mathcal{U}_*(\rho) = U\,\rho\, U^\dagger\,. \tag{26}$$

Remark. In infinite dimensions, there exist reversible dynamics that cannot be represented as conjugation by a unitary on a Hilbert space. The first place this arises is for infinite tensor products of qubits (spin chains). We will not encounter this subtlety in this course.

1.8.4. *The Schrödinger equation.* If the unitary $U(t)$ depends on a time parameter t in a *time-homogeneous* fashion, meaning

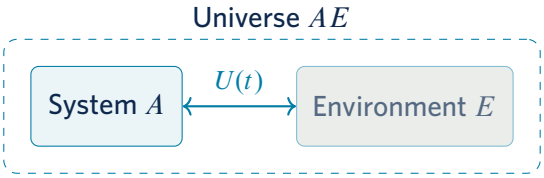
$$U(t+s) = U(t)\,U(s)\,, \qquad U(0) = \mathbb{I}\,, \tag{27}$$

then $U(t)$ forms a one-parameter group and satisfies the **Schrödinger equation**:

$$\boxed{i\,\frac{\mathrm{d}}{\mathrm{d}t}\,U(t) = H\,U(t)}\,, \tag{28}$$

where $H = H^\dagger$ is the *Hamiltonian*, a self-adjoint operator generating the evolution. The solution is $U(t) = e^{-iHt}$.

1.9. Open quantum systems. Is the Schrödinger equation the whole story? *Yes and no.* The aim of this course is to study the dynamics of *composite* quantum systems, and this leads us far beyond simple unitary evolution.



The picture is as follows. Our system A (an atom in a trap, a mode of light, a qubit) is embedded in a larger system AE that includes the environment E. The *universe* AE undergoes reversible (unitary) evolution $U(t)$. Our task: understand what this evolution looks like when *restricted to* A.

Irreversibility from ignoring subsystems

Even though the global evolution $U(t)$ is unitary and reversible, the effective evolution on subsystem A alone is, in general, *irreversible*. Irreversibility arises from tracing out (ignoring) the environment.

This is precisely how thermodynamics works: a system interacts with a reservoir; the joint evolution is unitary; but when we focus on the system alone, we see irreversible behaviour—dissipation, decoherence, thermalisation.

The Lindblad equation

Is there a "Schrödinger equation just for A"? Sometimes yes—under a Markov approximation, the effective evolution of A obeys a **Lindblad equation** (or master equation). Deriving and understanding this equation will be one of the main results of this course.

It used to be that open quantum systems were treated as an annoying detail to be left to a PhD student. This is no longer the case: dissipative dynamics, arising from interactions with a noisy environment, can actually be *useful*—they can prepare desired states and perform computations.

1.10. Quantum channels. We now axiomatically characterise the most general evolution in quantum mechanics: the *quantum channel*. Two features distinguish channels from the reversible evolutions of Section 1.8.3:

(a) The input and output systems may *differ*: a channel can map observables on a system B to observables on a system A, where A and B need not be the same.
(b) In addition to positivity, we require *complete positivity*.

Definition 1.4 (Quantum channel)**.** A *quantum channel* (in the Heisenberg picture) is a linear map

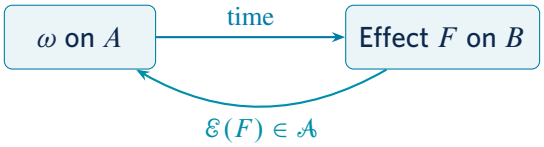
$$\mathcal{E}\colon \mathcal{B} \to \mathcal{A}$$

satisfying:

(1) **Unitality (probability preservation):** $\mathcal{E}(\mathbb{I}_B) = \mathbb{I}_A$.
(2) **Complete positivity:** for every $n \in \mathbb{N}$, the extended map $\mathcal{E} \otimes \mathrm{id}_n\colon \mathcal{B} \otimes M_n(\mathbb{C}) \to \mathcal{A} \otimes M_n(\mathbb{C})$ is positive:

$$X \geq 0 \implies (\mathcal{E} \otimes \mathrm{id}_n)(X) \geq 0\,. \tag{29}$$

1.10.1. *Why does $\mathcal{E}$ map $\mathcal{B} \to \mathcal{A}$?* In the Heisenberg picture, the channel acts on observables in the *reverse* direction to the flow of time:



If we prepare state ω on A and, after the evolution, measure effect F on B, then $\mathcal{E}(F)$ is the effective observable on A that gives the same statistics: $\omega(\mathcal{E}(F)) = p(F)$.

1.10.2. *Why complete positivity?* Ordinary positivity ($E \geq 0 \Rightarrow \mathcal{E}(E) \geq 0$) is not enough. Suppose we perform our evolution on system A while an ancillary system N sits idle in a neighbouring laboratory. The joint evolution is $\mathcal{E} \otimes \mathrm{id}_N$. If we then perform a *correlated* measurement on the joint system AN, we need the result to be a legal (positive) effect. This requires $\mathcal{E} \otimes \mathrm{id}_N$ to be positive for *every* ancillary system—which is precisely complete positivity.

Example 1.5 (The transpose map)**.** The map $T\colon A \mapsto A^\top$ (transpose in a fixed basis) is:

- linear,
- unital: $T(\mathbb{I}) = \mathbb{I}^\top = \mathbb{I}$,
- positive: $A \geq 0 \Rightarrow A^\top \geq 0$ (since eigenvalues are preserved by transposition).

However, T is *not* completely positive. The extended map $T \otimes \mathrm{id}_n$ fails to be positive: it maps certain positive elements of $M_n(\mathbb{C}) \otimes M_n(\mathbb{C})$ to operators with negative eigenvalues. Therefore, the transpose is **not** a valid quantum channel.

Exercise 1.2. Let $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ be a maximally entangled state of two qubits. Show that $(T \otimes \mathrm{id})(|\Phi^+\rangle\langle\Phi^+|)$ has a negative eigenvalue, thereby proving that the transpose map is not completely positive.

Hint: Compute the matrix explicitly and find its eigenvalues.

2. Complete positivity and dilation theory

In the previous lecture we defined quantum channels axiomatically: a channel $\mathcal{E}\colon \mathcal{B} \to \mathcal{A}$ must be unital and completely positive. We now ask: *can every channel be built from simple, physically transparent operations?*

2.1. Constructive approach to quantum operations.

We enumerate three elementary operations that should certainly be allowed in quantum mechanics.

2.1.1. *Adjoining an ancilla.* Given a system A in state ω, we may bring in an additional system B in a fixed state σ:

$$\omega \longmapsto \omega \otimes \sigma\,. \tag{30}$$

In the Schrödinger picture, this takes a state of A to a state of AB.

2.1.2. *Unitary evolution.* A closed system undergoes unitary dynamics:

$$\rho \longmapsto U\rho U^\dagger\,, \qquad U^\dagger U = \mathbb{I}\,. \tag{31}$$

2.1.3. *Reduction to a subsystem.* If we have a composite system AB and decide to ignore B (never measure anything on B), the effective state of A is obtained by partial trace:

$$\rho_{AB} \longmapsto \mathrm{tr}_B(\rho_{AB})\,. \tag{32}$$

Building arbitrary operations

By composing these three elementary operations, we can build remarkably complex processes. For example: start with a qubit A; adjoin an ancilla B in state σ; apply a joint unitary U on AB; adjoin a third system C in state σ_C; apply a different unitary V on BC; finally discard A, leaving an effective state of BC. Each step is one of the three allowed operations.

2.2. The Stinespring dilation theorem. The remarkable fact is that, for finite-dimensional systems, the axiomatic and constructive approaches are *equivalent*.

Theorem 2.1 (Stinespring dilation)**.** *Let $\mathcal{E}\colon M_N(\mathbb{C}) \to M_M(\mathbb{C})$ be a completely positive, unital map (i.e. a quantum channel in the Heisenberg picture). Then there exists a positive integer L and an isometry $V\colon \mathbb{C}^M \to \mathbb{C}^N \otimes \mathbb{C}^L$ (satisfying $V^\dagger V = \mathbb{I}_M$) such that*

$$\mathcal{E}(X) = V^\dagger (X \otimes \mathbb{I}_L) V \qquad \text{for all } X \in M_N(\mathbb{C}) . \tag{33}$$

The integer L can be chosen so that $L \leq M^2 N^2$.

In the Schrödinger picture, the dual map $\mathcal{E}_*$ acts on states as:

$$\mathcal{E}_*(\rho) = \mathrm{tr}_L\big(V \rho V^\dagger\big) . \tag{34}$$

This is precisely a composition of the three allowed operations:

$$\boxed{\rho} \xrightarrow{\text{adjoin}} \boxed{\rho \otimes |0\rangle\langle 0|} \xrightarrow{\text{unitary}} \boxed{U(\rho \otimes |0\rangle\langle 0|)U^\dagger} \xrightarrow{\text{reduce}} \boxed{\mathrm{tr}_L(\cdots)}$$

Remark. The Stinespring theorem is only valid for *finite-dimensional* systems. For infinite-dimensional systems, there exist completely positive, unital maps that cannot be written in the form (33). This means there may be legal quantum evolutions that do not arise from "system + environment + unitary"—a fact with implications for, e.g., black hole information loss.

3. Quantum many-particle systems

We now turn to modeling the *environment* in our system–environment picture. Real environments—the electromagnetic field, phonons in a solid, a thermal bath—consist of many (possibly indefinitely many) particles. We need mathematical machinery to handle this.

3.1. Fock space.

3.1.1. *Motivation.* We already know how to compose n quantum systems with Hilbert spaces $\mathcal{H}_{A_1}, \ldots, \mathcal{H}_{A_n}$: the joint Hilbert space is

$$\mathcal{H}_{A_1} \otimes \mathcal{H}_{A_2} \otimes \cdots \otimes \mathcal{H}_{A_n} . \tag{35}$$

But many physical systems—a mode of the electromagnetic field, Cooper pairs in a superconductor—have an *indefinite* number of particles: particles can be created and destroyed. For such systems, we need a Hilbert space that accommodates *any* number of particles simultaneously.

3.1.2. *Definition.* Let $\mathcal{H}$ be the single-particle Hilbert space. The **Fock space** (for distinguishable particles) is

$$\mathcal{F}(\mathcal{H}) = \bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n} \,, \tag{36}$$

where $\mathcal{H}^{\otimes 0} = \mathbb{C}$ (the "vacuum sector") and $\mathcal{H}^{\otimes n} = \underbrace{\mathcal{H} \otimes \cdots \otimes \mathcal{H}}_{n}$.

A vector $\Psi \in \mathcal{F}(\mathcal{H})$ is a sequence

$$\Psi = (\alpha_0,\ |\alpha_1\rangle,\ |\alpha_2\rangle,\ \ldots)\,, \qquad \alpha_0 \in \mathbb{C},\ \ |\alpha_n\rangle \in \mathcal{H}^{\otimes n}\,, \tag{37}$$

with the requirement that $\sum_n \|\alpha_n\|^2 < \infty$.

Remark. One might ask: why a direct sum rather than an infinite tensor product? The infinite tensor product $\mathcal{H}^{\otimes\infty}$ is strictly larger and contains states with no physical interpretation (e.g. superpositions involving infinitely many particles in incompatible ways). Fock space is the *smallest* Hilbert space that assigns a vector to every state we expect to exist.

The distinguished vector $|\Omega\rangle = (1, 0, 0, \ldots) \in \mathcal{F}(\mathcal{H})$ is the **vacuum state**: zero particles.

3.2. Identical particles and symmetrisation.

For *identical* particles (photons, electrons, …), swapping two particles must leave the state unchanged (up to a sign). We enforce this by projecting onto the appropriate symmetry sector.

3.2.1. *The symmetric group and permutation operators.* The **symmetric group** S_n is the group of all permutations of $\{1, 2, \ldots, n\}$. Each permutation $\pi \in S_n$ acts on product vectors by reordering the factors:

$$U_\pi\left(|v_1\rangle \otimes \cdots \otimes |v_n\rangle\right) = |v_{\pi^{-1}(1)}\rangle \otimes \cdots \otimes |v_{\pi^{-1}(n)}\rangle\,. \tag{38}$$

Define the **symmetrisation** and **antisymmetrisation** projectors:

$$P_n^{\pm} = \frac{1}{n!} \sum_{\pi \in S_n} \varepsilon_{\pm}(\pi)\, U_\pi\,, \tag{39}$$

where $\varepsilon_+(\pi) = 1$ for all π (bosons) and $\varepsilon_-(\pi) = \mathrm{sgn}(\pi)$ is the sign of the permutation (fermions).

3.2.2. *Bose and Fermi Fock space.*

Definition 3.1 (Bose/Fermi Fock space)**.** The **Bose Fock space** and **Fermi Fock space** are

$$\mathcal{F}_{\pm}(\mathcal{H}) = \bigoplus_{n=0}^{\infty} P_n^{\pm}\, \mathcal{H}^{\otimes n} \,. \tag{40}$$

In this course, we focus almost exclusively on *bosons* ($\mathcal{F}_+$), since our environments will typically be electromagnetic fields (photons). Everything we develop, however, carries over to fermions with only sign changes.

Example 3.1 (Single-mode Fock space)**.** If $\mathcal{H} = \mathbb{C}$ (a single mode of the electromagnetic field, e.g. the lowest mode of an optical cavity), then $\mathcal{H}^{\otimes n} \cong \mathbb{C}$ for every n, and Fock space becomes

$$\mathcal{F}_+(\mathbb{C}) = \bigoplus_{n=0}^{\infty} \mathbb{C} \cong \ell^2(\mathbb{N})\,. \tag{41}$$

This is the Hilbert space of a single quantum harmonic oscillator, spanned by number states $|0\rangle, |1\rangle, |2\rangle, \ldots$ The space is infinite-dimensional even though the single-particle space is one-dimensional—there can be any number of photons in the mode.

Example 3.2 (Multi-mode cavity)**.** If a cavity supports k modes (e.g. two polarisations of the lowest frequency), then $\mathcal{H} = \mathbb{C}^k$ and Fock space $\mathcal{F}_+(\mathbb{C}^k)$ describes an indefinite number of photons, each of which can occupy any of the k modes.

3.3. Creation and annihilation operators. Given a single-particle state $|f\rangle \in \mathcal{H}$, we define operators on the (distinguishable-particle) Fock space $\mathcal{F}(\mathcal{H})$ that add or remove a particle in state $|f\rangle$.

Definition 3.2 (Annihilation and creation operators)**.** The **annihilation operator** $a(f)$ and **creation operator** $a^\dagger(f)$ act on the n-particle sector as:

$$a(f)\,|\alpha_n\rangle = \sqrt{n}\,\left(\langle f| \otimes \mathbb{I}^{\otimes(n-1)}\right)|\alpha_n\rangle \in \mathcal{H}^{\otimes(n-1)}\,, \tag{42}$$

$$a^\dagger(f)\,|\alpha_n\rangle = \sqrt{n+1}\;P^{\pm}_{n+1}\left(|f\rangle \otimes |\alpha_n\rangle\right) \in \mathcal{H}^{\otimes(n+1)}\,, \tag{43}$$

and $a(f)\,|\Omega\rangle = 0$ (one cannot annihilate from the vacuum).

When restricted to the Bose or Fermi Fock space via the symmetrisation projectors $P^{\pm}_n$, these operators satisfy the fundamental **canonical commutation/anticommutation relations**:

$$[a(f),\, a^\dagger(g)]_\mp = \langle f|g\rangle\,\mathbb{I}\quad,\qquad [a(f),\, a(g)]_\mp = 0\,, \tag{44}$$

where $[\cdot,\cdot]_-$ is the commutator (bosons) and $[\cdot,\cdot]_+$ is the anticommutator (fermions).

The number operator

For a single mode ($\mathcal{H} = \mathbb{C}$, $|f\rangle = 1$), writing $a := a(1)$, the **number operator** $\hat{n} = a^\dagger a$ satisfies $\hat{n}|n\rangle = n|n\rangle$, where $|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle$ are the familiar number states of the quantum harmonic oscillator.

Remark. The notation $a(f)$ emphasises that the annihilation operator depends *linearly* on the single-particle state f, while $a^\dagger(f)$ depends *antilinearly* on f. In much of the physics literature, one fixes a mode basis $\{e_k\}$ and writes $a_k := a(e_k)$, recovering the familiar mode-indexed ladder operators.

3.4. Single-mode Fock space and the harmonic oscillator. When the single-particle Hilbert space is $\mathcal{H} = \mathbb{C}$ (a single mode of the electromagnetic field), Bose Fock space simplifies dramatically. Since $\mathbb{C}^{\otimes n} \cong \mathbb{C}$ for every n and the symmetrisation projector acts as the identity, we obtain

$$\mathcal{F}_+(\mathbb{C}) = \bigoplus_{n=0}^{\infty} \mathbb{C} \cong \ell^2(\mathbb{N})\,. \tag{45}$$

This is the Hilbert space of a single *quantum harmonic oscillator*, spanned by **Fock states** (number states) $|0\rangle, |1\rangle, |2\rangle, \ldots$ The creation and annihilation operators reduce to a single pair $a^\dagger$, a satisfying

$$[a,\, a^\dagger] = \mathbb{I}\,, \tag{46}$$

with the standard relations

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle\,,\qquad a|n\rangle = \sqrt{n}\,|n-1\rangle\,,\qquad a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle\,,\qquad a^\dagger a\,|n\rangle = n|n\rangle\,. \tag{47}$$

Definition 3.3 (Continuous quantum variable)**.** A *continuous quantum variable* (or *mode*) is a quantum system whose Hilbert space is the single-mode Fock space $\mathcal{F}_+(\mathbb{C}) \cong \ell^2(\mathbb{N})$.

This is the relevant description for a single mode of light in an optical cavity, a single phonon mode, or any quantum harmonic oscillator. In this course, both our system and its environment will typically be modelled as collections of continuous quantum variables.

4. Continuous variables

We now develop the mathematical toolkit for working with continuous quantum variables. The key challenge: states and operators on the infinite-dimensional space $\ell^2(\mathbb{N})$ require infinitely many parameters to specify. We need compact, physically meaningful ways to represent them.

4.1. Displacement operators.

Definition 4.1 (Displacement operator)**.** For $\alpha \in \mathbb{C}$, the **displacement operator** is

$$D(\alpha) = \exp\big(\alpha\, a^\dagger - \alpha^*\, a\big) \quad . \tag{48}$$

Since the exponent is i times a hermitian operator, $D(\alpha)$ is *unitary*: $D(\alpha)^\dagger = D(\alpha)^{-1} = D(-\alpha)$.

Why displacement operators?

Two reasons to prefer $D(\alpha)$ over the bare operators $a, a^\dagger$:

- **Mathematical:** a and $a^\dagger$ are unbounded; $D(\alpha)$ is bounded (unitary). We can recover a by differentiating $D(\alpha)$ with respect to α at $\alpha = 0$.
- **Physical:** displacement operators correspond to linear optics transformations (beam splitters, phase shifters). You can buy a device that implements $D(\alpha)$; you cannot buy a device that implements a.

Exercise 4.1. Using the Baker–Campbell–Hausdorff formula, show that

$$D(\alpha) = e^{-|\alpha|^2/2}\, e^{\alpha\, a^\dagger}\, e^{-\alpha^*\, a} \,. \tag{49}$$

The displacement operators satisfy the **composition rule**:

$$D(\alpha)\, D(\beta) = e^{(\alpha\beta^* - \alpha^*\beta)/2}\, D(\alpha + \beta) \,. \tag{50}$$

4.2. Coherent states.

Definition 4.2 (Coherent state)**.** The **coherent state** $|\alpha\rangle$ is obtained by displacing the vacuum:

$$|\alpha\rangle = D(\alpha)|0\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}}\, |n\rangle \,. \tag{51}$$

Coherent states are the states produced by lasers and are central to quantum optics. They are eigenstates of the annihilation operator: $a|\alpha\rangle = \alpha|\alpha\rangle$.

4.2.1. *Overcompleteness.* Coherent states form an *overcomplete* set: they span the Hilbert space, but are not orthogonal. The **resolution of the identity** reads

$$\frac{1}{\pi}\int_{\mathbb{C}} |\alpha\rangle\langle\alpha|\, \mathrm{d}^2\alpha = \mathbb{I} \quad , \qquad \mathrm{d}^2\alpha = \mathrm{d}(\mathrm{Re}\,\alpha)\, \mathrm{d}(\mathrm{Im}\,\alpha)\,. \tag{52}$$

Using this, any state $|\psi\rangle$ can be expanded as

$$|\psi\rangle = \frac{1}{\pi}\int_{\mathbb{C}} f(\alpha)\, |\alpha\rangle\, \mathrm{d}^2\alpha\,, \qquad f(\alpha) = \langle\alpha|\psi\rangle\,.$$

Knowing the function $f(\alpha)$ is equivalent to knowing the state. This is the simplest example of a *quasi-distribution* function.

Remark. Because the coherent states are overcomplete, the representation of a state in terms of $f(\alpha)$ is *not unique*—different functions can yield the same state. This redundancy is both a blessing (one can choose a convenient representative) and a curse (proving equality of two representations requires care).

4.3. Characteristic functions. For *mixed* states (density operators on an infinite-dimensional space), we need a way to specify operators—not just vectors—in terms of functions on $\mathbb{C}$.

4.3.1. *Orthogonality of displacement operators.* The displacement operators satisfy the trace orthogonality relation

$$\mathrm{tr}\big(D(\alpha)^\dagger\, D(\beta)\big) = \pi\, \delta^{(2)}(\alpha-\beta)\,, \tag{53}$$

where $\delta^{(2)}$ is the two-dimensional Dirac delta. As a consequence, any bounded operator A can be expanded as

$$A = \frac{1}{\pi}\int_{\mathbb{C}} \mathrm{tr}\big(A\, D(\alpha)^\dagger\big)\, D(\alpha)\, \mathrm{d}^2\alpha \quad . \tag{54}$$

4.3.2. *The characteristic function.*

Definition 4.3 (Characteristic function)**.** The **characteristic function** of a density operator ρ is

$$\chi_\rho(\alpha) = \mathrm{tr}\big(\rho\, D(\alpha)\big) \quad , \qquad \alpha \in \mathbb{C}\,. \tag{55}$$

Properties of the characteristic function

(i) **Normalisation:** $\chi_\rho(0) = \mathrm{tr}(\rho) = 1$.
(ii) **Hermiticity constraint:** $\chi_\rho(\alpha)^* = \chi_\rho(-\alpha)$.
(iii) **Determines ρ:** the density operator can be reconstructed from its characteristic function via

$$\rho = \frac{1}{\pi}\int_{\mathbb{C}} \chi_\rho(\alpha)\, D(-\alpha)\, \mathrm{d}^2\alpha\,. \tag{56}$$

(iv) **Experimentally accessible:** $\chi_\rho(\alpha)$ can be estimated from homodyne and heterodyne detection data.
(v) **Related to the Wigner function:** writing $\alpha = (q+ip)/\sqrt{2}$, the Fourier transform of χ_ρ yields the Wigner function (see next lecture).

4.4. Multiple modes. For n distinguishable modes, the Hilbert space is

$$\mathcal{F}_+(\mathbb{C})^{\otimes n} \cong \ell^2(\mathbb{N})^{\otimes n}\,, \tag{57}$$

with independent creation and annihilation operators $a_j^\dagger, a_j$ for $j = 1, \dots, n$ satisfying $[a_j, a_k^\dagger] = \delta_{jk}\mathbb{I}$.

The multi-mode displacement operator and coherent state are simply tensor products:

$$D(\boldsymbol{\alpha}) = D(\alpha_1) \otimes \cdots \otimes D(\alpha_n)\,, \qquad \boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n) \in \mathbb{C}^n\,, \tag{58}$$

$$|\boldsymbol{\alpha}\rangle = |\alpha_1\rangle \otimes \cdots \otimes |\alpha_n\rangle\,. \tag{59}$$

The overcompleteness relation generalises to

$$\frac{1}{\pi^n}\int_{\mathbb{C}^n} |\boldsymbol{\alpha}\rangle\langle\boldsymbol{\alpha}|\, \mathrm{d}^{2n}\boldsymbol{\alpha} = \mathbb{I}\,, \tag{60}$$

and the characteristic function of an n-mode state ρ is $\chi_\rho(\boldsymbol{\alpha}) = \mathrm{tr}(\rho\, D(\boldsymbol{\alpha}))$.

Remark. Multi-mode coherent states are always product states. The richer class of *Gaussian states*—which includes squeezed and entangled states—will be the subject of the next lecture.

4.5. Quadrature operators. So far we parameterised displacement operators using complex numbers $\boldsymbol{\alpha} \in \mathbb{C}^n$. It is standard in the Gaussian-state literature to work instead with their real and imaginary parts.

Definition 4.4 (Quadrature operators)**.** For a single mode with annihilation operator a, define the **quadrature operators**

$$\hat{q} = \frac{a + a^\dagger}{\sqrt{2}}\,, \qquad \hat{p} = \frac{a - a^\dagger}{i\sqrt{2}}\,. \tag{61}$$

Both are hermitian and satisfy $[\hat{q}, \hat{p}] = i\mathbb{I}$.

For n modes we collect the $2n$ quadrature operators into a vector

$$\hat{\boldsymbol{R}} = (\hat{q}_1, \dots, \hat{q}_n, \hat{p}_1, \dots, \hat{p}_n)^\top\,. \tag{62}$$

The commutation relations are encoded by the **symplectic matrix**

$$[\hat{R}_j, \hat{R}_k] = i\,\sigma_{jk}\,\mathbb{I}\,, \qquad \sigma = \begin{pmatrix} 0 & \mathbb{I}_n \\ -\mathbb{I}_n & 0 \end{pmatrix}\,. \tag{63}$$

4.5.1. *Displacement operator in quadrature notation.* For $\boldsymbol{\xi} = (x_1, \dots, x_n, p_1, \dots, p_n)^\top \in \mathbb{R}^{2n}$ the displacement operator becomes

$$D(\boldsymbol{\xi}) = \exp\!\big(i\,\boldsymbol{\xi}^\top \sigma\, \hat{\boldsymbol{R}}\big)\,, \tag{64}$$

with $\alpha_j = (x_j + i\,p_j)/\sqrt{2}$ connecting to the complex parameterisation of Lecture 4. The characteristic function in these variables is

$$\chi_\rho(\boldsymbol{\xi}) = \mathrm{tr}\big(\rho\, D(\boldsymbol{\xi})\big)\,, \qquad \boldsymbol{\xi} \in \mathbb{R}^{2n}\,. \tag{65}$$

4.6. Gaussian states.

Definition 4.5 (Gaussian state)**.** An n-mode state ρ is a **Gaussian state** if its characteristic function is a Gaussian in $\boldsymbol{\xi}$:

$$\boxed{\chi_\rho(\boldsymbol{\xi}) = \exp\!\Big(-\tfrac{1}{4}\,\boldsymbol{\xi}^\top \gamma\, \boldsymbol{\xi} - i\,\boldsymbol{m}^\top \sigma\, \boldsymbol{\xi}\Big)}\,, \tag{66}$$

where $\boldsymbol{m} \in \mathbb{R}^{2n}$ is the **displacement vector** and γ is a real $2n \times 2n$ symmetric matrix called the **covariance matrix**.

The absence of a constant term is fixed by normalisation: $\chi_\rho(\mathbf{0}) = \mathrm{tr}(\rho) = 1$.
The displacement vector records the **first moments**

$$m_j = \langle \hat{R}_j \rangle = \mathrm{tr}(\rho\, \hat{R}_j)\,, \tag{67}$$

and the covariance matrix records the symmetrised **second moments**

$$\gamma_{jk} = \langle \{\hat{R}_j - m_j,\ \hat{R}_k - m_k\} \rangle\,, \tag{68}$$

where $\{A, B\} = AB + BA$ is the anticommutator.

Why Gaussian states are remarkable

A Gaussian state of n modes lives in an infinite-dimensional Hilbert space, yet is completely determined by $\boldsymbol{m}$ ($2n$ reals) and γ ($2n(2n+1)/2$ entries). All higher moments follow, as for a classical Gaussian, making polynomial expectation values computable on paper.

Remark. Superpositions of Gaussian states are *not* generally Gaussian. A superposition of the vacuum and a coherent state produces a Schrödinger-cat-like state, which is non-Gaussian.

4.7. Examples of Gaussian states.

Example 4.1 (Vacuum state)**.** The n-mode vacuum $\rho = |\mathbf{0}\rangle\langle\mathbf{0}|$ is Gaussian with

$$\boldsymbol{m} = \mathbf{0}\,, \qquad \gamma = \mathbb{I}_{2n}\,. \tag{69}$$

It is a minimum-uncertainty state: $(\Delta q)^2 = (\Delta p)^2 = \frac{1}{2}$.

Example 4.2 (Coherent states)**.** Every coherent state $|\boldsymbol{\alpha}\rangle$ is Gaussian. For a single mode with $\alpha = (x_0 + i\, p_0)/\sqrt{2}$:

$$\boldsymbol{m} = \begin{pmatrix} x_0 \\ p_0 \end{pmatrix}, \qquad \gamma = \mathbb{I}_2\,. \tag{70}$$

The covariance matrix equals that of the vacuum; only the displacement shifts, consistent with $|\alpha\rangle = D(\alpha)|0\rangle$.

4.8. Legitimate covariance matrices. Not every symmetric matrix γ corresponds to a physical quantum state. Positivity of ρ imposes a constraint.

Covariance matrix condition

A real symmetric matrix $\gamma \in \mathbb{R}^{2n\times 2n}$ is the covariance matrix of a quantum state if and only if

$$\boxed{\gamma + i\sigma \geq 0}\,, \tag{71}$$

where the inequality is positive semi-definiteness as a hermitian matrix. This is the continuous-variable uncertainty principle.

In particular, $\gamma > 0$ (strictly positive definite): zero eigenvalues are forbidden.

Proof (necessity). For any $\boldsymbol{c} \in \mathbb{C}^{2n}$, define $A = \sum_j c_j\, (\hat{R}_j - m_j)$. Since $\rho \geq 0$:

$$0 \leq \mathrm{tr}(\rho\, A^\dagger A) = \sum_{j,k} \bar{c}_j\, c_k\ \mathrm{tr}\big(\rho\, (\hat{R}_j - m_j)(\hat{R}_k - m_k)\big) = \tfrac{1}{2} \sum_{j,k} \bar{c}_j\, c_k\, (\gamma_{jk} + i\sigma_{jk})\,. \tag{72}$$

Since this holds for all $\boldsymbol{c}$, we conclude $\gamma + i\sigma \geq 0$. □

Remark. The converse also holds: every $(\boldsymbol{m}, \gamma)$ satisfying (71) gives a legitimate Gaussian state. Constructively, for any such γ there exists a system of harmonic oscillators whose *thermal state* has that covariance matrix, since thermal states of quadratic Hamiltonians are always Gaussian.

4.9. Bipartite Gaussian states and reduced states. Consider two parties, Alice (A) and Bob (B), each holding n modes (joint system: $2n$ modes), described by a displacement vector $\boldsymbol{m}_{AB}$ and a $4n \times 4n$ covariance matrix γ_{AB}.

Proposition 4.1 (Reduced states of Gaussian states)**.** *If ρ_{AB} is Gaussian, then the reduced state $\rho_A = \mathrm{tr}_B(\rho_{AB})$ is also Gaussian, with*

$$\boldsymbol{m}_A = (\boldsymbol{m}_{AB})_A\,, \qquad \gamma_A = (\gamma_{AB})_{AA}\,, \tag{73}$$

i.e. one simply extracts the sub-vector and sub-matrix corresponding to Alice's modes.

Proof. The characteristic function of the reduced state is

$$\chi_{\rho_A}(\boldsymbol{\xi}_A) = \mathrm{tr}_{AB}\big(\rho_{AB}\,(D(\boldsymbol{\xi}_A) \otimes \mathbb{I}_B)\big) = \chi_{\rho_{AB}}(\boldsymbol{\xi}_A, \boldsymbol{0}_B)\,,$$

using $D(\boldsymbol{0}) = \mathbb{I}$. Setting $\boldsymbol{\xi}_B = \boldsymbol{0}$ in the Gaussian form (66) yields a Gaussian in $\boldsymbol{\xi}_A$ with the stated parameters. □

Remark. Reducing Gaussian states is trivial (extract a sub-matrix), but the reverse—*extending*—is subtler. Trivial extensions by tensoring always exist, but *pure* Gaussian extensions (purifications) require a non-trivial construction; they always exist and will appear in later lectures. For tripartite systems ABC, it is possible that ρ_{AB} and ρ_{BC} are both Gaussian yet no global ρ_{ABC} is consistent with both marginals—a manifestation of entanglement monogamy.

4.10. Example: the two-mode squeezed state. The **two-mode squeezed state** is the canonical entangled Gaussian state, produced by parametric down-conversion in a nonlinear crystal. For two modes A, B with $\hat{\boldsymbol{R}} = (\hat{q}_A, \hat{q}_B, \hat{p}_A, \hat{p}_B)^\top$ and squeezing parameter $r \geq 0$:

$$\boldsymbol{m} = \boldsymbol{0}\,, \qquad \gamma = \begin{pmatrix} \cosh 2r & \sinh 2r & 0 & 0 \\ \sinh 2r & \cosh 2r & 0 & 0 \\ 0 & 0 & \cosh 2r & -\sinh 2r \\ 0 & 0 & -\sinh 2r & \cosh 2r \end{pmatrix}. \tag{74}$$

At $r = 0$ this is the two-mode vacuum ($\gamma = \mathbb{I}_4$). In the Fock basis the corresponding pure state is

$$|\psi_r\rangle = \frac{1}{\cosh r} \sum_{n=0}^{\infty} (\tanh r)^n\, |n\rangle_A |n\rangle_B\,. \tag{75}$$

For large r the coefficients decay slowly, so $|\psi_r\rangle$ resembles a maximally entangled state up to a photon-number cutoff—one of the primary entanglement sources in continuous-variable quantum information.

Exercise 4.2. Verify Proposition 4.1 for the two-mode squeezed state: (a) extract the sub-matrix of (74) for mode A; (b) compute $\rho_A = \mathrm{tr}_B(|\psi_r\rangle\langle\psi_r|)$ directly from (75). In both cases, confirm $\gamma_A = \cosh(2r)\,\mathbb{I}_2$ (a thermal state).

4.11. Quadratic Hamiltonians and Gaussian models. To model the environment E in our system–environment picture, we need exactly solvable models for bosonic fields. *Quadratic* (Gaussian) Hamiltonians fill this role: they are general enough to capture physics from phonon propagation in solids to relativistic quantum field theory, yet they can be diagonalised using only linear algebra.

Definition 4.6 (Gaussian (quasi-free) Hamiltonian)**.** Given n modes with canonical operators $\hat{x}_1, \ldots, \hat{x}_n, \hat{p}_1, \ldots, \hat{p}_n$ satisfying the CCR $[\hat{x}_j, \hat{p}_k] = i\,\delta_{jk}\mathbb{I}$, the **Gaussian Hamiltonian** is

$$\hat{H} = \tfrac{1}{2}\sum_{j,k=1}^{n} [V_x]_{jk}\,\hat{x}_j\,\hat{x}_k + \tfrac{1}{2}\sum_{j,k=1}^{n} [V_p]_{jk}\,\hat{p}_j\,\hat{p}_k \quad , \tag{76}$$

where $V_x, V_p \in \mathbb{R}^{n\times n}$ are symmetric, positive-definite matrices encoding the potential and kinetic energy, respectively.

Scope of the Gaussian model

Despite having only quadratic terms, this Hamiltonian covers: V_x, V_p diagonal $\Rightarrow$ independent oscillators; V_x off-diagonal, $V_p = \mathbb{I} \Rightarrow$ phonons in solids; both off-diagonal $\Rightarrow$ the Klein–Gordon field on a lattice. The mode number n can be arbitrarily large; all results below remain valid.

4.11.1. *Compact notation.* Introduce the column vector of $2n$ canonical operators

$$\hat{\boldsymbol{r}} = (\hat{x}_1, \ldots, \hat{x}_n, \hat{p}_1, \ldots, \hat{p}_n)^\top \tag{77}$$

and the $2n \times 2n$ Hamiltonian matrix

$$h = \begin{pmatrix} V_x & 0 \\ 0 & V_p \end{pmatrix} . \tag{78}$$

Then the Hamiltonian takes the compact form

$$\hat{H} = \tfrac{1}{2}\,\hat{\boldsymbol{r}}^\top h\,\hat{\boldsymbol{r}} . \tag{79}$$

In terms of ladder operators $\hat{b}_j = (\hat{x}_j + i\,\hat{p}_j)/\sqrt{2}$, the Hamiltonian contains number-conserving terms $(\hat{b}_j^\dagger \hat{b}_k)$ and *squeezing* terms $(\hat{b}_j \hat{b}_k, \hat{b}_j^\dagger \hat{b}_k^\dagger)$ that do not conserve particle number.

4.12. Symplectic transformations. The strategy for diagonalising (79) is to make a linear change of variables on the operators that preserves the CCR.

Definition 4.7 (Symplectic form and symplectic group)**.** The $2n \times 2n$ **symplectic form** is

$$\sigma = \begin{pmatrix} 0 & \mathbb{I}_n \\ -\mathbb{I}_n & 0 \end{pmatrix} . \tag{80}$$

The canonical commutation relations in matrix form read

$$[\hat{r}_j, \hat{r}_k] = i\,\sigma_{jk}\,\mathbb{I} . \tag{81}$$

A real $2n \times 2n$ matrix S is **symplectic** if it preserves the symplectic form:

$$S\,\sigma\,S^\top = \sigma \quad . \tag{82}$$

The set of all such matrices forms the **real symplectic group** $\mathrm{Sp}(2n, \mathbb{R})$.

Defining $\hat{\boldsymbol{r}}' = S\,\hat{\boldsymbol{r}}$ with $S \in \mathrm{Sp}(2n, \mathbb{R})$ preserves the CCR. Our task reduces to finding an S that brings h to diagonal form.

4.13. Williamson's theorem.

Theorem 4.2 (Williamson's normal form)**.** *Let M be a $2n\times 2n$ real, symmetric, positive-definite matrix. Then there exists a symplectic matrix $S \in \mathrm{Sp}(2n,\mathbb{R})$ such that*

$$S\,M\,S^{\top} = \mathrm{diag}(\mu_1,\dots,\mu_n,\ \mu_1,\dots,\mu_n)\ , \tag{83}$$

*where the **symplectic eigenvalues** $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n > 0$ are uniquely determined by M.*

Remark. The symplectic eigenvalues can be found without constructing S: they are the positive eigenvalues of the matrix $i\sigma M$, or equivalently the positive square roots of the eigenvalues of $-(\sigma M)^2$.

4.13.1. *Constructing S for the Gaussian Hamiltonian.* For $h = \mathrm{diag}(V_x, V_p)$, an explicit symplectic diagonaliser can be constructed. Define the $n\times n$ matrix

$$G = V_p^{1/2}\,V_x\,V_p^{1/2}\,, \tag{84}$$

which is symmetric and positive definite. Let Q be an orthogonal matrix diagonalising G:

$$Q^{\top}G\,Q = D^2 = \mathrm{diag}(\mu_1^2,\dots,\mu_n^2)\,, \tag{85}$$

where $\mu_j > 0$ are the symplectic eigenvalues of h.

Explicit symplectic diagonaliser

The matrix

$$S = \begin{pmatrix} D^{1/2}\,Q^{\top}V_x^{-1/2} & 0 \\ 0 & D^{-1/2}\,Q^{\top}V_p^{-1/2} \end{pmatrix} \tag{86}$$

is symplectic and satisfies $S\,h\,S^{\top} = \mathrm{diag}(\mu_1,\dots,\mu_n,\ \mu_1,\dots,\mu_n)$.

Exercise 4.3. Verify that (86) satisfies $S\sigma S^{\top} = \sigma$ and $ShS^{\top} = \mathrm{diag}(\mu_1,\dots,\mu_n,\mu_1,\dots,\mu_n)$.

When is this easy?

In non-relativistic physics $V_p = \mathbb{I}$, so $G = V_x$. For translation-invariant systems V_x is circulant and diagonalisable by the discrete Fourier transform, making the construction analytic.

4.14. Diagonalised form and normal modes. Defining new canonical variables $\hat{\boldsymbol{r}}' = S\,\hat{\boldsymbol{r}}$ (which still obey the CCR), the Hamiltonian becomes

$$\hat{H} = \sum_{j=1}^{n}\frac{\mu_j}{2}\big(\hat{x}_j'^2 + \hat{p}_j'^2\big) = \sum_{j=1}^{n}\mu_j\Big(\hat{a}_j^{\dagger}\hat{a}_j + \tfrac{1}{2}\Big)\,, \tag{87}$$

where $\hat{a}_j = (\hat{x}_j' + i\,\hat{p}_j')/\sqrt{2}$ are the **normal-mode** annihilation operators. This is a collection of n independent harmonic oscillators with frequencies $\mu_1,\dots,\mu_n$.

Reduction to independent oscillators

Every Gaussian Hamiltonian (76) is unitarily equivalent to a sum of independent harmonic oscillators—justifying the name *quasi-free*.

4.15. Transformation of the characteristic function. The symplectic change of variables acts simply on the characteristic function: if $\chi_\rho(\boldsymbol{\xi}) = \mathrm{tr}(\rho\, D(\boldsymbol{\xi}))$, then

$$\chi_\rho(\boldsymbol{\xi}) = \chi_{\rho'}\big((S^{-1})^\top \boldsymbol{\xi}\big)\,, \tag{88}$$

where ρ' is the state in the normal-mode basis. This implies simple transformation rules for the moments:

$$\boldsymbol{m} = S^{-1}\,\boldsymbol{m}'\,, \qquad \gamma = S^{-1}\,\gamma'\,(S^{-1})^\top \quad , \tag{89}$$

where $\boldsymbol{m} = \langle \hat{\boldsymbol{r}} \rangle$ and $\gamma_{jk} = \frac{1}{2}\langle\{\Delta\hat{r}_j,\, \Delta\hat{r}_k\}\rangle$ with $\Delta\hat{r}_j = \hat{r}_j - m_j$.

4.16. The thermal state of a Gaussian model. Since $\hat{H}$ is a sum of independent oscillators in the normal-mode basis, the thermal state $\rho_\beta = e^{-\beta\hat{H}}/Z$ factorises:

$$\rho_\beta = \bigotimes_{j=1}^{n} \frac{e^{-\beta\mu_j(\hat{a}_j^\dagger \hat{a}_j + 1/2)}}{Z_j}\,, \qquad Z_j = \frac{1}{2\sinh(\beta\mu_j/2)}\,. \tag{90}$$

Each factor is Gaussian, so the full thermal state is Gaussian with first moments $\boldsymbol{m} = 0$ (by symmetry) and covariance matrix:

Covariance matrix of the thermal state

In the normal-mode variables, the covariance matrix is diagonal:

$$\gamma' = \mathrm{diag}\big(\bar{n}_1 + \tfrac{1}{2}, \ldots, \bar{n}_n + \tfrac{1}{2},\, \bar{n}_1 + \tfrac{1}{2}, \ldots, \bar{n}_n + \tfrac{1}{2}\big)\,, \tag{91}$$

where $\bar{n}_j = (e^{\beta\mu_j} - 1)^{-1}$ is the Bose–Einstein mean occupation number. In the original variables, the covariance matrix is

$$\gamma_\beta = S^{-1}\,\gamma'\,(S^{-1})^\top \quad . \tag{92}$$

Remark. Proving that a single-mode thermal state has a Gaussian characteristic function requires summing Laguerre polynomials over Fock states and applying a generating-function identity. The result: $\chi(\xi) \propto \exp\big(-(\bar{n} + \frac{1}{2})|\xi|^2\big)$.

Exercise 4.4. For n independent oscillators with $V_x = \mathrm{diag}(\omega_1^2, \ldots, \omega_n^2)$ and $V_p = \mathbb{I}$, show that:

(a) The symplectic eigenvalues are $\mu_j = \omega_j$.
(b) The covariance matrix in the original variables is $\gamma_\beta = \mathrm{diag}\big((\bar{n}_1 + \frac{1}{2})/\omega_1, \ldots, (\bar{n}_n + \frac{1}{2})/\omega_n,\, (\bar{n}_1 + \frac{1}{2})\omega_1, \ldots, (\bar{n}_n + \frac{1}{2})\omega_n\big)$.

Why this result matters

The covariance matrix (92) is the starting point for modelling the environment. Combined with Wick's theorem it determines all correlation functions of the thermal field, and its entries will later set the decay rates in the Lindblad equation.

5. Lindblad equations

Over the past lectures we assembled all the ingredients needed to describe a system S interacting with an environment E: completely positive maps, Fock space, Gaussian models. Our goal now is to derive an *equation of motion* for the system alone—one that does not require tracking every degree of freedom of the environment.

5.1. Setup: system plus environment. The total Hilbert space is $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$, where $\mathcal{H}_S$ is typically finite-dimensional and $\mathcal{H}_E$ models the environment—often an infinite collection of harmonic oscillators. The joint Hamiltonian is

$$H = H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E + V \quad , \tag{93}$$

with free part $H_0 = H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E$ and interaction V. Any Hamiltonian on $\mathcal{H}_S \otimes \mathcal{H}_E$ admits such a decomposition.

Why the full description fails

The exact reduced state $\rho_S(t) = \mathrm{tr}_E\big(U(t)\,(\rho_S(0) \otimes \rho_E)\,U(t)^\dagger\big)$ requires the joint unitary $U(t) = e^{-iHt}$, which couples the few degrees of freedom of S to the (infinitely many) degrees of freedom of E. We need a description involving only S.

5.2. Five assumptions toward a master equation.

5.2.1. *Assumption 1: initially decoupled state.* At $t = 0$ the system and environment are in a product state:

$$\rho(0) = \rho_S(0) \otimes \rho_E \quad , \tag{94}$$

justified when S has been insulated from E by design or the coupling is weak. If relaxed, the reduced dynamics need not even be completely positive.

5.2.2. *Assumption 2: weak coupling.* V is a *small* perturbation of H_0, controlled by a dimensionless coupling constant (e.g. the fine-structure constant $\alpha \approx 1/137$).

Remark. One may always decompose the interaction-picture coupling as $\tilde{V}(t) = \tilde{V}_S(t) \otimes \mathbb{I}_E + \tilde{V}_{SE}(t)$ with $\mathrm{tr}_E\,(\tilde{V}_{SE}(t)\,\rho_E) = 0$, by absorbing any system-only part into a redefined H_S. This ensures the first-order contribution vanishes after tracing over E.

5.3. The interaction picture. Define $U_0(t) = e^{-iH_0 t}$ and set

$$\tilde{V}(t) = U_0(t)^\dagger V\, U_0(t)\,, \qquad \tilde{\rho}(t) = U_0(t)^\dagger \rho(t)\, U_0(t)\,. \tag{95}$$

The equation of motion in the rotating frame involves only the interaction:

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{\rho}(t) = -i\big[\tilde{V}(t),\, \tilde{\rho}(t)\big] \quad . \tag{$\star$}$$

Integrate from 0 to t and substitute back into ($\star$) to obtain an equation correct to second order in V:

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{\rho}(t) = -i\big[\tilde{V}(t),\, \tilde{\rho}(0)\big] - \int_0^t \mathrm{d}t_1\, \big[\tilde{V}(t),\, \big[\tilde{V}(t_1),\, \tilde{\rho}(t_1)\big]\big]\,. \tag{96}$$

Iterating further produces the **Dyson series**; we stop at second order because the first-order term will vanish after tracing over E and all leading physical effects (dissipation, decoherence) appear at this order.

5.4. Tracing out the environment. Taking tr_E of (96) and using $\mathrm{tr}_E(\tilde{V}_{SE}(t)\,\rho_E) = 0$ to kill the first-order term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{\rho}_S(t) = -\int_0^t \mathrm{d}t_1\ \mathrm{tr}_E\Big[\tilde{V}(t),\ \big[\tilde{V}(t_1),\ \tilde{\rho}(t_1)\big]\Big]. \tag{97}$$

This is exact to second order but still involves the full system–environment state inside the integral.

5.5. The Born approximation (Assumption 3). Replace the full state inside the integral by a product:

$$\boxed{\tilde{\rho}(t_1)\ \approx\ \tilde{\rho}_S(t_1)\otimes\rho_E}\ . \tag{98}$$

Why neglecting entanglement is justified

The Born approximation looks drastic: it discards all system–environment entanglement. But we *never interrogate the environment*. A photon emitted by the system flies away at the speed of light and never returns; since we cannot perform a Bell test between S and E, the correlations are invisible to any system-only measurement. The approximation breaks down if the environment can "talk back" (e.g. photons bouncing between mirrors).

Substituting into (97):

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{\rho}_S(t) = -\int_0^t \mathrm{d}t_1\ \mathrm{tr}_E\Big[\tilde{V}(t),\ \big[\tilde{V}(t_1),\ \tilde{\rho}_S(t_1)\otimes\rho_E\big]\Big]. \tag{99}$$

5.6. The Markov approximation (Assumptions 4 and 5). **Assumption 4** (dense bath spectrum). The environment has many closely spaced energy levels in every interval ΔE.

Continuous spectra and irreversibility

Only baths with absolutely continuous spectrum exhibit irreversible information loss—a consequence of the *RAGE theorem*. Discrete spectra lead to Poincaré recurrences.

Environmental correlations therefore decay on a time scale τ_E much shorter than the system relaxation time, permitting the replacement $\tilde{\rho}_S(t_1) \approx \tilde{\rho}_S(t)$ inside the integral.

Assumption 5 (Markov). The bath spectrum is truly continuous, so we may extend the integration:

$$\int_0^t \mathrm{d}t_1\ (\cdots)\ \longrightarrow\ \int_0^\infty \mathrm{d}\tau\ (\cdots)\,, \qquad \tau = t - t_1\,. \tag{100}$$

Born–Markov master equation

Combining Assumptions 1–5, the reduced system state satisfies

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\tilde{\rho}_S(t) = -\int_0^\infty \mathrm{d}\tau\ \mathrm{tr}_E\Big[\tilde{V}(t),\ \big[\tilde{V}(t-\tau),\ \tilde{\rho}_S(t)\otimes\rho_E\big]\Big]}\ . \tag{101}$$

Remark. The intermediate equation before Assumption 5 (with $\tilde{\rho}_S(t)$ but the original limits) is the **Redfield equation**—time-local but with time-dependent coefficients.

5.7. Example: two-level atom in the vacuum field.

Spontaneous emission

Spontaneous decay of an excited atom was explained by Dirac–Wigner–Weisskopf shortly after the formulation of quantum mechanics—one of the earliest triumphs of quantum field theory.

5.7.1. *The model.* The environment is a free bosonic field, $H_E = \sum_k \omega_k\, b_k^\dagger b_k$. The system is a two-level atom with ground state $|g\rangle$, excited state $|e\rangle$, and transition frequency ω_A:

$$H_S = \omega_A\, |e\rangle\langle e|\,. \tag{102}$$

The dipole coupling is

$$V = \sum_k \big(g_k\, \sigma_+ \otimes b_k + g_k^*\, \sigma_- \otimes b_k^\dagger\big)\,, \tag{103}$$

where $\sigma_+ = |e\rangle\langle g|$, $\sigma_- = \sigma_+^\dagger$, and the coupling constants g_k encode dipole matrix elements and mode structure.

5.7.2. *Rotating-wave approximation.* In the interaction picture the coupling acquires phases $e^{\pm i(\omega_A-\omega_k)t}$ (slowly rotating) and $e^{\pm i(\omega_A+\omega_k)t}$ (fast counter-rotating). Since $\omega_A \sim 10^{15}\,\mathrm{s}^{-1}$, the fast terms self-average and can be dropped (**rotating-wave approximation**, RWA):

$$\tilde{V}(t) \approx \sum_k \big(g_k\, \sigma_+ b_k\, e^{i(\omega_A-\omega_k)t} + g_k^*\, \sigma_- b_k^\dagger\, e^{-i(\omega_A-\omega_k)t}\big)\,. \tag{104}$$

5.7.3. *Preview of the master equation.* Setting $\rho_E = |0\rangle\langle 0|$ (vacuum) and carrying out the Born–Markov procedure (detailed next lecture), one obtains:

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_S = -i[H_S,\, \rho_S] + \gamma\Big(\sigma_-\rho_S\,\sigma_+ - \tfrac{1}{2}\sigma_+\sigma_-\rho_S - \tfrac{1}{2}\rho_S\,\sigma_+\sigma_-\Big)\,, \tag{105}$$

where $\gamma > 0$ is the spontaneous-emission rate.

Irreversibility from unitarity

The joint dynamics is unitary, yet the atom alone decays irreversibly. The resolution: one excitation can flow into a continuum of field modes, but the reverse—all modes conspiring to re-excite the atom—has vanishing probability.

Exercise 5.1. Verify that (105) preserves the trace and positivity of ρ_S:

(a) Show $\frac{\mathrm{d}}{\mathrm{d}t}\,\mathrm{tr}(\rho_S) = 0$.
(b) Writing ρ_S in the $\{|e\rangle, |g\rangle\}$ basis, derive the equations of motion for the matrix elements and check that $\rho_{ee}(t) = \rho_{ee}(0)\, e^{-\gamma t}$.

5.8. Recap: the radiative-damping model. We continue the derivation begun in Lecture 7. The setting is a two-level atom (system S) coupled to a bosonic field (environment E) via dipole interaction. In the interaction picture, after the **rotating-wave approximation** (RWA), the interaction Hamiltonian is

$$\hat{V}_{\mathrm{int}}(t) = \underbrace{\sum_k g_k\, \sigma_+\, \hat{b}_k\, e^{i(\omega_a-\omega_k)t}}_{V_+(t)} + \underbrace{\sum_k g_k^*\, \sigma_-\, \hat{b}_k^\dagger\, e^{-i(\omega_a-\omega_k)t}}_{V_-(t)}\,, \tag{106}$$

where $\sigma_\pm$ are the atomic raising/lowering operators, $\hat{b}_k$, $\hat{b}_k^\dagger$ are the field mode operators with frequencies ω_k, and g_k are the coupling constants. The initial state of the environment is taken to be the vacuum: $\rho_E(0) = |0\rangle\langle 0|$.

Physical picture of radiative damping

With the field in the vacuum, no photons impinge on the atom. An excited atom can only lose energy: the term $V_-(t)$ creates a photon in mode k while flipping the atom downward. The created photon flies away and (by the Markov assumption) never returns. Hence an initial excited state $|1\rangle$ decays irreversibly to the ground state $|0\rangle$.

5.9. Deriving the master equation.

5.9.1. *The Born-approximation equation.* In the Born approximation the equation of motion for the system density operator is

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_S(t) = -\int_0^t \mathrm{d}t_1 \ \mathrm{tr}_E\big[\hat{V}_{\mathrm{int}}(t),\, [\hat{V}_{\mathrm{int}}(t_1),\, \rho_S(t_1)\otimes|0\rangle\langle 0|]\big]\,. \tag{107}$$

Expanding the double commutator with $V_{\mathrm{int}} = V_+ + V_-$ produces $4\times 4 = 16$ terms.

5.9.2. *Vacuum simplification.* The key identity that eliminates most terms is:

$$\mathrm{tr}_E\big((\mathbb{I}_S\otimes A_E)\,\rho_S\otimes|0\rangle\langle 0|\,(\mathbb{I}_S\otimes B_E)\big) = \langle 0|B_E\,A_E|0\rangle\,\rho_S\,, \tag{108}$$

together with $\hat{b}_k|0\rangle = 0$ for all modes k. Since $V_+(t)$ contains annihilation operators acting on the field, any term where V_+ stands adjacent to $|0\rangle$ vanishes. A non-zero contribution requires equally many V_+ and V_- factors, reducing the 16 terms to just 4.

Remark. The surviving terms pair creation and annihilation operators so that the field expectation value $\langle 0|\hat{b}_k\,\hat{b}_l^\dagger|0\rangle = \delta_{kl}$ produces a Kronecker delta, collapsing double sums over modes to single sums.

5.9.3. *Continuum limit and the bath correlation function.* Replacing the discrete mode sum by an integral introduces the **density of states** $\rho(\omega)$:

$$\sum_k |g_k|^2\, f(\omega_k) \longrightarrow \int_0^\infty \mathrm{d}\omega\ \rho(\omega)\,|g(\omega)|^2\, f(\omega)\,. \tag{109}$$

The bath correlation function that appears in (107) is

$$\gamma(\tau) = \int_0^\infty \mathrm{d}\omega\ \rho(\omega)\,|g(\omega)|^2\, e^{i(\omega_a-\omega)\tau}\,. \tag{110}$$

Markov approximation

If the product $\rho(\omega)|g(\omega)|^2$ is *slowly varying* around $\omega \approx \omega_a$, then its Fourier transform $\gamma(\tau)$ is sharply peaked near $\tau = 0$, acting effectively as a delta function under the time integral in (107). This converts the integro-differential (non-Markovian) equation into a memoryless (Markovian) equation of motion.

5.9.4. *The Lindblad equation for radiative damping.* Decomposing $\int_0^\infty \gamma(\tau)\,\mathrm{d}\tau$ into real and imaginary parts,

$$\int_0^\infty \gamma(\tau)\,\mathrm{d}\tau = \frac{\gamma}{2} + i\,\delta\omega_a\,, \tag{111}$$

and combining all terms, the master equation becomes

$$\frac{\mathrm{d}\rho_S}{\mathrm{d}t} = -i\left[\tfrac{\omega_a+\delta\omega_a}{2}\,\sigma_z,\ \rho_S\right] + \gamma\,\mathcal{D}[\sigma_-]\rho_S\,, \tag{112}$$

where the **dissipator** (or **Lindblad superoperator**) is defined by

$$\mathcal{D}[L]\rho := L\,\rho\,L^\dagger - \tfrac{1}{2}\,L^\dagger L\,\rho - \tfrac{1}{2}\,\rho\,L^\dagger L\,. \tag{113}$$

Definition 5.1 (Lindblad equation)**.** An equation of the form $\dot\rho = -i\,[H,\rho] + \sum_j \gamma_j\,\mathcal{D}[L_j]\rho$, with $\gamma_j \geq 0$ and $H = H^\dagger$, is called a **Lindblad (or GKSL) master equation**. The operators L_j are called **Lindblad operators** (or **jump operators**).

Remark. The frequency shift $\delta\omega_a$ is the **Lamb shift**: the coupling to the vacuum field renormalises the atomic transition frequency. For most purposes it simply redefines ω_a and can be absorbed by moving to a suitable rotating frame.

Complete positivity

Any Lindblad equation generates a completely positive, trace-preserving (CPTP) semigroup $\rho(t) = e^{\mathcal{L}t}\rho(0)$. Conversely, for finite-dimensional systems, every continuous Markovian CPTP evolution is generated by a Lindblad equation. This is the content of the *Gorini–Kossakowski–Sudarshan–Lindblad theorem*.

5.10. Solving the radiative-damping equation. Setting $\delta\omega_a = 0$ (absorbing the Lamb shift into the rotating frame), we solve the Lindblad equation (112) with $L = \sigma_- = |0\rangle\langle 1|$. Parameterise the qubit density operator as

$$\rho_S(t) = \begin{pmatrix} 1-p(t) & x(t) \\ x^*(t) & p(t) \end{pmatrix}, \tag{114}$$

where $p(t)$ is the excited-state population and $x(t)$ encodes the coherence. Substituting into the Lindblad equation yields two decoupled equations:

$$\dot p = -\gamma\,p\,, \qquad \dot x = -\tfrac{\gamma}{2}\,x\,. \tag{115}$$

Solution: radiative damping of a two-level atom

The qubit density operator at time t is

$$\rho_S(t) = \begin{pmatrix} 1-p(0)\,e^{-\gamma t} & x(0)\,e^{-\gamma t/2} \\ x^*(0)\,e^{-\gamma t/2} & p(0)\,e^{-\gamma t} \end{pmatrix}. \tag{116}$$

- The excited-state population decays at rate γ.
- The coherences decay at rate $\gamma/2$ (half the population decay rate).
- The **steady state** is the ground state: $\rho_{\mathrm{ss}} = \lim_{t\to\infty}\rho_S(t) = |0\rangle\langle 0|$.

Dynamics on the Bloch sphere

On the Bloch sphere, every initial state spirals inward toward the north pole $|0\rangle$. The excited state $|1\rangle$ (south pole) moves straight up along the z-axis; superposition states follow curved trajectories as the coherence decays faster than the population. After a characteristic time $\sim 1/\gamma$ the atom is, with high probability, in its ground state.

5.11. Cavity emission. As a second example, we consider a single cavity mode (a harmonic oscillator with annihilation operator a and frequency ω_c) that leaks through an imperfect mirror into external radiation modes.

The leaky cavity

A photon inside the cavity bounces between the mirrors. Each reflection at the imperfect mirror has a small probability of transmission; the photon eventually escapes and is lost to the external field. This is the bosonic analogue of radiative damping.

The total Hamiltonian is quadratic (field–field coupling):

$$H = \omega_c\, a^\dagger a + \sum_k \omega_k\, \hat{b}_k^\dagger \hat{b}_k + \sum_k \big(g_k\, a^\dagger \hat{b}_k + g_k^*\, a\, \hat{b}_k^\dagger\big)\,. \tag{117}$$

The derivation follows the same steps as for radiative damping with the substitution $\sigma_+ \to a^\dagger$, $\sigma_- \to a$.

5.11.1. *Vacuum environment.* If the external field is in the vacuum, we obtain

$$\frac{\mathrm{d}\rho_S}{\mathrm{d}t} = -i\,[\omega_c\, a^\dagger a,\ \rho_S] + \kappa\, \mathcal{D}[a]\,\rho_S\,, \tag{118}$$

where $\kappa > 0$ is the cavity decay rate.

5.11.2. *Thermal environment.* If the external field is in a thermal state at frequency ω_c with mean photon number $\bar{n} = (e^{\beta\omega_c} - 1)^{-1}$, an additional term accounts for incoherent photon injection:

$$\boxed{\frac{\mathrm{d}\rho_S}{\mathrm{d}t} = -i\,[\omega_c\, a^\dagger a,\ \rho_S] + \kappa(\bar{n}+1)\,\mathcal{D}[a]\,\rho_S + \kappa\bar{n}\,\mathcal{D}[a^\dagger]\,\rho_S}\,. \tag{119}$$

5.12. Solving the cavity equation via the characteristic function. For the vacuum-environment case (118) (working in the frame rotating at ω_c), we translate the master equation into an equation for the characteristic function $\chi(\psi,t) = \mathrm{tr}\big(\rho_S(t)\, D(\psi)\big)$. Using the identities

$$a^\dagger\, D(\psi) = \Big(\frac{\partial}{\partial\psi} + \frac{\psi^*}{2}\Big) D(\psi)\,, \qquad a\, D(\psi) = \Big(-\frac{\partial}{\partial\psi^*} + \frac{\psi}{2}\Big) D(\psi)\,, \tag{120}$$

the master equation becomes a first-order PDE:

$$\frac{\partial\chi}{\partial t} = -\frac{\kappa}{2}\Big(\psi\,\frac{\partial\chi}{\partial\psi} + \psi^*\frac{\partial\chi}{\partial\psi^*} + |\psi|^2\,\chi\Big)\,. \tag{121}$$

This is a Fokker–Planck-type equation whose solution preserves the Gaussian form. Making the ansatz $\chi(\psi,t) = \exp\big(-\frac{1}{2}K(t)|\psi|^2\big)$ for an initial Gaussian state with parameter $K(0)$, one finds

$$K(t) = 1 + \big(K(0) - 1\big)\, e^{-\kappa t}\,. \tag{122}$$

Steady state of a leaky cavity

As $t \to \infty$, $K(t) \to 1$ and $\chi(\psi) \to e^{-|\psi|^2/2}$, which is the characteristic function of the vacuum state. Every initial state of the cavity decays to the vacuum at rate κ.

Exercise 5.2. Complete the derivation of the remaining three terms in the radiative damping master equation (107). Use the vacuum identities and verify that the result matches (112).

Exercise 5.3. Starting from the cavity Hamiltonian (117) with a thermal environment, carry out the Born–Markov derivation to obtain (119). Identify where the additional $\mathcal{D}[a^\dagger]$ term arises.

Exercise 5.4. Derive equation (121) from (118) using the displacement-operator identities (120). Verify that the Gaussian ansatz yields (122).

6. Quasi-free fermion systems

In the preceding lectures we developed the theory of Gaussian bosonic models: quadratic Hamiltonians, symplectic diagonalisation, and covariance matrices. We now turn to the *fermionic* counterpart. Fermion systems arise whenever we model conducting materials: a quantum dot—an artificial atom acting as a cavity for electrons—embedded in a resistor is the prototypical example. Just as for bosons, every quadratic fermion Hamiltonian can be diagonalised exactly, and its ground state is determined by a covariance matrix.

6.1. General quadratic fermion Hamiltonian. Consider n spinless fermion modes with annihilation operators $\hat{f}_1, \dots, \hat{f}_n$ obeying the canonical anticommutation relations (CAR)

$$\{\hat{f}_j, \hat{f}_k^\dagger\} = \delta_{jk}\,\mathbb{I}\,, \qquad \{\hat{f}_j, \hat{f}_k\} = 0\,. \tag{123}$$

Definition 6.1 (Quasi-free fermion Hamiltonian)**.** The most general quadratic Hamiltonian for n fermion modes is

$$\hat{H} = \sum_{j,k=1}^{n} A_{jk}\,\hat{f}_j^\dagger \hat{f}_k + \tfrac{1}{2}\sum_{j,k=1}^{n}\left(B_{jk}\,\hat{f}_j^\dagger \hat{f}_k^\dagger + B_{jk}^*\,\hat{f}_k \hat{f}_j\right)\,, \tag{124}$$

where $A, B \in \mathbb{R}^{n\times n}$. Hermiticity requires $A = A^\top$ (symmetric) and $B = -B^\top$ (antisymmetric).

Remark. The pairing terms involving B appear to violate fermion-number parity superselection rules but arise as *effective* terms in superconducting systems without creating superpositions of even and odd fermion number in the full theory.

Introducing the $2n$-dimensional Nambu vector $\hat{\boldsymbol{F}} = (\hat{f}_1, \dots, \hat{f}_n, \hat{f}_1^\dagger, \dots, \hat{f}_n^\dagger)^\top$, the Hamiltonian takes the compact form

$$\hat{H} = \tfrac{1}{2}\,\hat{\boldsymbol{F}}^\dagger\,\mathcal{H}\,\hat{\boldsymbol{F}} + \text{const}\,, \qquad \mathcal{H} = \begin{pmatrix} A & B \\ -B & -A \end{pmatrix}\,. \tag{125}$$

6.2. Majorana operators. Just as bosonic quadrature operators are hermitian combinations of a and $a^\dagger$, we build hermitian fermion operators.

Definition 6.2 (Majorana operators)**.** For each mode $j = 1, \dots, n$, define

$$\hat{c}_{2j-1} = \frac{\hat{f}_j + \hat{f}_j^\dagger}{\sqrt{2}}\,, \qquad \hat{c}_{2j} = \frac{\hat{f}_j - \hat{f}_j^\dagger}{i\sqrt{2}}\,. \tag{126}$$

These $2n$ operators are hermitian, $\hat{c}_k^\dagger = \hat{c}_k$, and satisfy

$$\{\hat{c}_j, \hat{c}_k\} = \delta_{jk}\,\mathbb{I} \quad . \tag{127}$$

Majorana fermions

Majorana operators are the fermion analogue of the quadrature operators $\hat{q}$, $\hat{p}$. Majorana zero modes at the ends of superconducting nanowires have attracted intense experimental interest as a potential platform for topological quantum computation.

Collecting these into $\hat{\boldsymbol{c}} = (\hat{c}_1, \ldots, \hat{c}_{2n})^\top$, the Hamiltonian becomes

$$\hat{H} = \frac{i}{2}\,\hat{\boldsymbol{c}}^\top V\,\hat{\boldsymbol{c}} + \text{const}\,, \tag{128}$$

where V is the $n \times n$ real matrix $V_{jk} = A_{jk} + B_{jk}$ (up to conventions for the block decomposition).

6.3. Diagonalisation via singular value decomposition. Unlike the bosonic case (symplectic transformations), the matrix V is in general neither symmetric nor antisymmetric. The appropriate tool is the **singular value decomposition** (SVD).

SVD diagonalisation of the fermion Hamiltonian

Every real $n \times n$ matrix V admits a decomposition

$$V = O_1\,\Sigma\,O_2^\top\,, \tag{129}$$

with $O_1, O_2 \in O(n)$ orthogonal and $\Sigma = \text{diag}(\sigma_1, \ldots, \sigma_n)$, $\sigma_j \geq 0$. The orthogonal matrices are found by diagonalising $VV^\top$ (giving O_1) and $V^\top V$ (giving O_2). Define new Majorana operators

$$\hat{\boldsymbol{c}}' = \begin{pmatrix} O_1 & 0 \\ 0 & O_2 \end{pmatrix} \hat{\boldsymbol{c}}\,, \tag{130}$$

which preserve the anticommutation relations (127), and revert to creation/annihilation operators $\hat{b}_j = (\hat{c}'_{2j-1} + i\,\hat{c}'_{2j})/\sqrt{2}$. The Hamiltonian then takes the diagonal form

$$\hat{H} = \sum_{j=1}^{n} \sigma_j \left(\hat{b}_j^\dagger \hat{b}_j - \tfrac{1}{2}\right) \quad . \tag{131}$$

Exercise 6.1. Verify that the transformation (130) preserves the Majorana anticommutation relations (127).

6.4. Spectrum and ground state. Each mode $\hat{b}_j$ is an independent **fermion oscillator** with a two-dimensional Hilbert space $\{|0\rangle_j, |1\rangle_j\}$. The full Hilbert space has dimension 2^n—finite, in contrast to the bosonic case.

Complete spectrum

The eigenvalues are labelled by occupation vectors $\boldsymbol{n} = (n_1, \ldots, n_n) \in \{0,1\}^n$:

$$E_{\boldsymbol{n}} = \sum_{j=1}^{n} \sigma_j \left(n_j - \tfrac{1}{2}\right). \tag{132}$$

The **ground state** $|\Omega\rangle = |0\rangle_1 \cdots |0\rangle_n$ (the $\hat{b}$-vacuum) has energy $E_0 = -\frac{1}{2}\sum_{j=1}^{n} \sigma_j$. It is unique when rank $V = n$; otherwise the ground-state degeneracy is $2^{n-\operatorname{rank} V}$.

The **spectral gap** is the smallest singular value: $\Delta = \sigma_{\min}(V)$.

The ground state is not empty

The $\hat{b}$-vacuum contains no quasiparticles, but generically contains many $\hat{f}$-fermions—this is the Fermi sea. The average fermion number in the original variables is

$$\langle \hat{N} \rangle_\Omega = \frac{n}{2} - \frac{1}{2}\operatorname{tr}(\bar{V}), \tag{133}$$

where $\bar{V} = O_1\, O_2^\top$ is the *spectral flattening* of V (SVD with all singular values set to 1). Depending on O_1 and O_2, this ranges from 0 (empty) to n (completely filled), reproducing the familiar picture of a conductor at some chemical potential.

6.5. Fermion covariance matrix. Gaussian fermion states are characterised by their covariance matrix in the Majorana basis, just as Gaussian boson states are characterised by their covariance matrix in the $(\hat{q}, \hat{p})$ variables.

Definition 6.3 (Fermion covariance matrix)**.** The **covariance matrix** of a state ρ of n fermion modes is the $2n \times 2n$ real antisymmetric matrix

$$\Gamma_{jk} = -i\, \langle [\hat{c}_j, \hat{c}_k] \rangle = -2i\, \langle \hat{c}_j \hat{c}_k \rangle + i\, \delta_{jk}. \tag{134}$$

From Γ one extracts all second moments of the original operators. For the ground state:

$$\langle \hat{f}_j^\dagger \hat{f}_k \rangle_\Omega = \frac{1}{2}\left(\delta_{jk} - [\bar{V}]_{jk}\right), \tag{135}$$

where $\bar{V} = O_1\, O_2^\top$ is the spectral flattening.

Bosons versus fermions

For bosons the Hilbert space is infinite-dimensional and the covariance matrix is real symmetric. For fermions the Hilbert space is 2^n-dimensional and the covariance matrix is real antisymmetric. In both cases, the Gaussian state is completely determined by its second moments.

6.6. Application: tunnelling through a quantum dot. We apply the quasi-free fermion formalism to a **quantum dot**—an artificial atom (single orbital, charging energy E_c) cooled to ~ 1 K and coupled by tunnel junctions to two fermion reservoirs (source and drain) held at chemical potentials μ_S and μ_D. The bias voltage $V_{\text{bias}} = (\mu_S - \mu_D)/e$ drives a current through the dot. This is the fermion analogue of an optical cavity coupled to input and output modes.

6.6.1. *The Hamiltonian.* The total Hamiltonian $\hat{H} = \hat{H}_0 + \hat{V}$ consists of a free part

$$\hat{H}_0 = E_c\, \hat{d}^\dagger \hat{d} + \sum_{\alpha \in \{S,D\}} \sum_k \varepsilon_k^{(\alpha)}\, \hat{f}_{k,\alpha}^\dagger \hat{f}_{k,\alpha}, \tag{136}$$

where $\hat{d}$ is the dot fermion operator, $\hat{f}_{k,\alpha}$ are reservoir operators with kinetic energies $\varepsilon_k^{(\alpha)} = \hbar^2 k^2/2m$, and a tunnelling interaction

$$\hat{V} = \sum_{\alpha\in\{S,D\}} \sum_k \left(t_{k,\alpha}\, \hat{d}^\dagger \hat{f}_{k,\alpha} + t^*_{k,\alpha}\, \hat{f}^\dagger_{k,\alpha} \hat{d}\right), \tag{137}$$

with small tunnelling matrix elements $t_{k,\alpha}$ (classically forbidden transitions).

Remark. The full Hamiltonian is quadratic and could be diagonalised exactly. However, just as for the optical cavity, we will instead derive a Born–Markov master equation for the dot alone in the next lecture, providing more physical insight into the transport process.

6.6.2. *Initial state and regime of validity.* At $t = 0$ the tunnel barriers are high and the state factorises:

$$\rho(0) = \rho_S \otimes \rho_{\text{dot}} \otimes \rho_D\,, \tag{138}$$

where ρ_S and ρ_D are Fermi-sea ground states at their respective chemical potentials (not empty states). The barriers are then lowered and the system evolves under $\hat{H}$.

When does the master equation work?

The Born–Markov approximation requires the tunnelling rates $\gamma_L, \gamma_R \propto |t_{k,\alpha}|^2$ to be small, so that fermions hop one-by-one through the dot. This high-resistance regime is realised only at very low temperatures (~ 1 K); at room temperature the sequential-tunnelling picture breaks down.

Exercise 6.2. Consider $n = 2$ fermion modes with $A = \left(\begin{smallmatrix} 1 & \epsilon \\ \epsilon & 2 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & \delta \\ -\delta & 0 \end{smallmatrix}\right)$ for small $\epsilon, \delta > 0$.

(a) Construct V and compute its singular values.
(b) Write down the ground-state energy and the spectral gap.
(c) Compute $\langle \hat{N} \rangle_\Omega$ in the ground state.

6.7. Quantum-dot master equation. We return to the fermionic system introduced in the previous lecture: a single quantum dot coupled to two external leads (source L and drain R) by the Hamiltonian

$$H = \underbrace{\sum_k \epsilon_k\, a_k^\dagger a_k + E_c\, c^\dagger c + \sum_p \epsilon_p\, d_p^\dagger d_p}_{H_0} + \underbrace{\sum_k T_k\, a_k^\dagger c + \sum_p T'_p\, c^\dagger d_p + \text{h.c.}}_{V}\,, \tag{139}$$

where a_k annihilates a fermion in mode k of the source, c is the annihilation operator for the single bound state of the dot (energy E_c), and d_p annihilates a fermion in mode p of the drain. An external voltage biases the leads so that electrons flow from source to drain through the dot.

Fermionic analogue of cavity decay

This model is the fermionic counterpart of the optical cavity coupled to input and output modes: electrons transit through the dot just as photons transit through a cavity.

6.7.1. *Interaction picture and reservoir correlators.* Going to the interaction picture with respect to H_0 and expanding the von Neumann equation to second order—exactly as in the Born–Markov derivation for cavity decay—the dynamics depend on two types of non-vanishing reservoir correlators:

$$C_n^{(L)}(\tau) = \sum_k |T_k|^2\, \bar{n}_k^{(L)}\, e^{-i(\epsilon_k - E_c)\tau}\,, \qquad C_a^{(L)}(\tau) = \sum_k |T_k|^2\, (1 - \bar{n}_k^{(L)})\, e^{i(\epsilon_k - E_c)\tau}\,, \tag{140}$$

with $\bar{n}_k^{(L)} = \langle a_k^\dagger a_k \rangle_\beta$ the thermal occupation of mode k in the source (analogous expressions hold for the drain). The anti-normally ordered correlator C_a is non-zero because of fermionic anti-commutation relations—absent in the bosonic case.

6.7.2. *Continuum limit and Markov approximation.* Replacing the discrete mode sum by an integral with density of states $\rho(\omega)$, only frequencies near E_c contribute appreciably (stationary-phase argument).

Fermi–Dirac occupation

The mean occupation of a fermionic mode at energy ω is

$$\bar{n}(\omega) = \frac{1}{e^{\beta(\omega-\omega_F)}+1}\,, \tag{141}$$

where ω_F is the Fermi energy. At low temperature, $\bar{n}(\omega) \to \Theta(\omega_F - \omega)$: all modes below the Fermi surface are filled.

The bias sets $E_c < \omega_F^{(L)}$ and $E_c > \omega_F^{(R)}$; at low temperature $\bar{n}^{(L)}(E_c) \approx 1$ and $\bar{n}^{(R)}(E_c) \approx 0$. The Markov approximation then yields effective coupling rates

$$\frac{\gamma_L}{2} = \pi\,\rho(E_c)\,|T(E_c)|^2\,\bar{n}^{(L)}(E_c)\,, \qquad \frac{\gamma_R}{2} = \pi\,\rho(E_c)\,|T'(E_c)|^2\,\big(1-\bar{n}^{(R)}(E_c)\big)\,. \tag{142}$$

6.7.3. *Master equation for the dot.* Following the same steps as for cavity decay, the master equation for the reduced state ρ of the dot ($\mathcal{H} \cong \mathbb{C}^2$) is

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \gamma_L\Big(c^\dagger \rho\, c - \tfrac{1}{2}\{c\,c^\dagger,\,\rho\}\Big) + \gamma_R\Big(c\,\rho\,c^\dagger - \tfrac{1}{2}\{c^\dagger c,\,\rho\}\Big)\,. \tag{143}$$

The first term describes electrons tunnelling *onto* the dot from the source (rate γ_L); the second describes electrons tunnelling *off* the dot into the drain (rate γ_R).

Remark. Since the dot has one bound state and the electron is spin-polarised, the fermionic superselection rule forbids superpositions of $|0\rangle$ and $|1\rangle$; the dynamics reduce to a single number.

6.7.4. *Solution: occupation dynamics.* Defining $\langle \hat{n} \rangle = \mathrm{tr}(c^\dagger c\,\rho)$ and using fermionic identities ($c^2 = 0$, $(c^\dagger c)^2 = c^\dagger c$), one obtains

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle \hat{n} \rangle = \gamma_L\big(1 - \langle \hat{n} \rangle\big) - \gamma_R\,\langle \hat{n} \rangle\,. \tag{144}$$

The steady-state occupation is $\langle \hat{n} \rangle_\infty = \gamma_L/(\gamma_L+\gamma_R)$, and the full solution with initial condition $\langle \hat{n} \rangle(0) = n_0$ reads

$$\langle \hat{n} \rangle(t) = \langle \hat{n} \rangle_\infty + \big(n_0 - \langle \hat{n} \rangle_\infty\big)\,e^{-(\gamma_L+\gamma_R)\,t}\,. \tag{145}$$

Regardless of its initial state, the dot relaxes exponentially to $\langle \hat{n} \rangle_\infty$ at rate $\gamma_L + \gamma_R$.

Exercise 6.3. Derive (143) by carrying out the Born–Markov procedure for (139), replacing the bosonic bath correlators with $\gamma_L/2$ and $\gamma_R/2$.

6.8. Lindblad structure: a general pattern. The master equations derived so far—two-level atom, cavity decay, quantum dot—all share the *Lindblad form*:

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -i\,[H_{\text{sys}},\,\rho] + \sum_j \gamma_j\,\mathcal{D}[L_j]\,\rho\,, \qquad \mathcal{D}[L]\,\rho = L\rho L^\dagger - \tfrac{1}{2}\{L^\dagger L,\,\rho\}\,. \tag{146}$$

Why always Lindblad form?

Complete positivity of the evolved state for all $t \geq 0$, under the Markov assumption, severely constrains the generator. The Lindblad theorem (forthcoming) shows that (146) is the *most general* such generator.

7. Classical noise and stochastic calculus

7.1. Repeated quantum measurements. We now present a derivation that provides perhaps the clearest physical intuition for the Lindblad structure. It arises from a model of *repeated quantum measurements* due to Caves and Milburn [**?**].

Caves–Milburn (1987)

In the Caves–Milburn model the Lindblad master equation arises *exactly*—not as an approximation—from repeatedly coupling a system to independent meters.

7.1.1. *The setup.* Consider a system S with Hamiltonian H_{sys} and n independent meters $M_1, \ldots, M_n$ (free particles). The total Hilbert space is

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_S \otimes \mathcal{H}_{M_1} \otimes \cdots \otimes \mathcal{H}_{M_n}\,. \tag{147}$$

The system evolves freely except at discrete times $t_j = j\tau$, when it interacts instantaneously with meter M_j via the **von Neumann measurement Hamiltonian**

$$\boxed{H_{\text{int}}^{(j)} = \kappa\,\hat{A} \otimes \hat{p}_j}\,, \tag{148}$$

where $\hat{A}$ is the system observable to be measured, $\hat{p}_j$ is the momentum of meter j, and κ controls the interaction strength.

7.1.2. *Von Neumann measurement prescription.*

Definition 7.1 (Von Neumann measurement model)**.** The meter is prepared in the position eigenstate $|x = 0\rangle_M$. The system–meter composite evolves under H_{int} for a time τ, producing the unitary $U = \exp(-i\kappa\tau\,\hat{A} \otimes \hat{p})$.

Expanding $|\psi\rangle = \sum_j c_j |\alpha_j\rangle$ in the eigenbasis of $\hat{A}$, the initial product state $|\Psi(0)\rangle = |\psi\rangle \otimes |x = 0\rangle_M$ evolves to

$$\boxed{|\Psi(\tau)\rangle = \sum_j c_j\,|\alpha_j\rangle \otimes |x = \kappa\tau\alpha_j\rangle_M}\,, \tag{149}$$

since $e^{-i\kappa\tau\alpha_j\hat{p}}$ is the translation operator shifting position by $\kappa\tau\alpha_j$. Each eigenvalue is now correlated with a distinct pointer reading.

Effective projective measurement

A subsequent projective measurement of the pointer position yields outcome $x = \kappa\tau\alpha_j$ with probability $|c_j|^2$, and the system collapses to $|\alpha_j\rangle$. This reproduces the textbook measurement postulate for $\hat{A}$.

7.1.3. *Repeated measurements and scaling regimes.* The full time-dependent Hamiltonian is

$$H(t) = H_{\text{sys}} + \kappa \sum_{j=1}^{n} \delta(t - j\tau)\, \hat{A} \otimes \hat{p}_j \,. \tag{150}$$

The physics depends on the scaling of n and κ:

- **Strong coupling** (κ large): each interaction is nearly projective. Increasing the measurement rate freezes the system in an eigenstate of $\hat{A}$—the **quantum Zeno effect**.
- **Weak coupling** ($\kappa \to 0$, n fixed): the meters decouple; the system evolves unitarily under H_{sys}.
- **Intermediate scaling** ($n \to \infty$, $\kappa \to 0$, $n\kappa^2$ fixed): the meters exert a finite back-action. This yields a Lindblad master equation and provides a transparent model for environmental decoherence.

The Goldilocks regime

Each individual measurement barely disturbs the system, but infinitely many weak measurements produce continuous, irreversible evolution—precisely how a thermal environment induces decoherence.

The derivation of the Lindblad equation from this repeated-measurement model will be carried out in the next lecture.

7.2. From repeated measurements to continuous evolution. We now carry out the programme announced in the previous lecture: starting from n discrete von Neumann measurements of position, we take $n \to \infty$ and $\tau \to 0$ simultaneously, obtaining a Lindblad master equation that models *continuous measurement*.

Decoherence as information flow

A system loses coherence when the environment gains information about it. Information gain builds up entanglement between system and environment. Continuous measurement provides a transparent model for this process.

7.2.1. *Hilbert space and Hamiltonian.* Recall the setup: the system is a continuous quantum variable with canonical operators $\hat{x}_S$, $\hat{p}_S$ satisfying $[\hat{x}_S, \hat{p}_S] = i$, and we introduce n independent meter systems with canonical pairs $(\hat{x}_j, \hat{p}_j)$. The total Hilbert space is

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_{M_1} \otimes \cdots \otimes \mathcal{H}_{M_n} \otimes \mathcal{H}_S \,, \qquad \mathcal{H}_{M_j} = L^2(\mathbb{R}) \,. \tag{151}$$

The time-dependent Hamiltonian modelling n instantaneous measurements at times $t_j = j\tau$ is

$$\boxed{H(t) = H_0 + \sum_{j=1}^{n} \delta(t - j\tau)\, \hat{x}_S \otimes \hat{p}_j} \,, \tag{152}$$

where H_0 is the free system Hamiltonian. Each meter is prepared in a Gaussian state of width $\sqrt{\sigma}$:

$$g_j(x_j) = \langle x_j | g_j \rangle = \frac{1}{(\pi\sigma)^{1/4}} \exp\Big(-\frac{x_j^2}{2\sigma}\Big), \tag{153}$$

so the total initial state is $\rho(0) = |g_1\rangle\langle g_1| \otimes \cdots \otimes |g_n\rangle\langle g_n| \otimes \rho_S$.

Remark. The parameter σ controls measurement resolution: a narrow Gaussian ($\sigma \to 0$) approximates a projective measurement; a wide one ($\sigma \to \infty$) yields almost no information.

7.3. Selective and non-selective state update. Write $\rho_S(t_j^-)$ and $\rho_S(t_j^+)$ for the system state just before and just after the j-th interaction. Integrating the Schrödinger equation across the delta-function impulse gives

$$\rho_S(t_j^+) = \mathrm{tr}_{M_j}\Big[U_j \left(|g_j\rangle\langle g_j| \otimes \rho_S(t_j^-)\right) U_j^\dagger\Big], \quad U_j = e^{-i\,\hat{x}_S \otimes \hat{p}_j}. \tag{154}$$

Selective measurement. If we measure the pointer and obtain outcome $\bar{x}_j$, the system is updated to the sub-normalised conditional state $\tilde{\rho}_S(t_j^+ \mid \bar{x}_j) = \hat{\Upsilon}(\bar{x}_j)\, \rho_S(t_j^-)\, \hat{\Upsilon}(\bar{x}_j)^\dagger$, where the **measurement operator** is

$$\hat{\Upsilon}(\bar{x}_j) = \frac{1}{(\pi\sigma)^{1/4}} \exp\Big(-\frac{(\hat{x}_S - \bar{x}_j)^2}{2\sigma}\Big). \tag{155}$$

The normalised posterior is $\rho_S(t_j^+ \mid \bar{x}_j) = \tilde{\rho}_S / p(\bar{x}_j)$ with $p(\bar{x}_j) = \mathrm{tr}[\hat{\Upsilon}(\bar{x}_j)^2\, \rho_S(t_j^-)]$.
Non-selective measurement. If the meter outcome is discarded (or controlled by the environment), we average over all outcomes:

$$\rho_S(t_j^+) = \int_{-\infty}^{\infty} \mathrm{d}\bar{x}_j\; \hat{\Upsilon}(\bar{x}_j)\; \rho_S(t_j^-)\; \hat{\Upsilon}(\bar{x}_j)^\dagger. \tag{156}$$

This defines a completely positive, trace-preserving map on ρ_S.

Measurement back-action

In quantum mechanics, information gain always entails disturbance. Even when the meter outcome is discarded, the interaction has irreversibly altered the system state—this is the physical origin of decoherence.

7.4. Full evolution and measurement trajectories. After j measurements with outcomes $\bar{x}_1, \ldots, \bar{x}_j$, the conditional system state is

$$\rho_S(t_j^+ \mid \bar{x}_1, \ldots, \bar{x}_j) = \frac{\hat{M}_j\; \rho_S(0)\; \hat{M}_j^\dagger}{p(\bar{x}_1, \ldots, \bar{x}_j)}, \tag{157}$$

where the time-ordered product is

$$\hat{M}_j = \hat{\Upsilon}(\bar{x}_j)\, U(\tau)\; \hat{\Upsilon}(\bar{x}_{j-1})\, U(\tau)\; \cdots\; \hat{\Upsilon}(\bar{x}_1)\, U(\tau), \tag{158}$$

with $U(\tau) = e^{-iH_0\tau}$, and $p(\bar{x}_1, \ldots, \bar{x}_j) = \mathrm{tr}[\hat{M}_j\, \rho_S(0)\, \hat{M}_j^\dagger]$. The sequence $(\bar{x}_1, \ldots, \bar{x}_n)$ is the **measurement trajectory**. Averaging over all trajectories yields the unconditional (non-selective) state

$$\rho_S(t_j^+) = \int \mathrm{d}\bar{x}_1 \cdots \mathrm{d}\bar{x}_j\; \hat{M}_j\; \rho_S(0)\; \hat{M}_j^\dagger. \tag{159}$$

7.5. The continuous limit. Fix a total observation time T and set $\tau = T/n$. As $n \to \infty$, both $\tau \to 0$ and the number of measurements grows without bound.

Gaussian integral identity

Evaluating (178) in closed form and expanding for large σ, one obtains the single-step update

$$\rho_S(t_j^+) = \rho_S(t_j^-) - \frac{1}{8\sigma}\big[\hat{x}_S, [\hat{x}_S, \rho_S(t_j^-)]\big] + O(1/\sigma^2)\,. \tag{160}$$

Combining with the free evolution $\rho_S(t_{j+1}^-) = U(\tau)\,\rho_S(t_j^+)\,U(\tau)^\dagger$ and expanding $U(\tau) = \mathbb{I} - iH_0\tau + O(\tau^2)$, the state increment over one period is

$$\rho_S(t_{j+1}^+) - \rho_S(t_j^+) = -i\,[H_0, \rho_S]\,\tau - \frac{1}{8\sigma}\big[\hat{x}_S, [\hat{x}_S, \rho_S]\big] + O(\tau^2)\,. \tag{161}$$

Dividing by τ and taking $n \to \infty$, the character of the limiting equation depends on how σ scales with τ:

(1) **Quantum Zeno regime** (σ fixed): the $1/\sigma$ term dominates; the system freezes in a position eigenstate.
(2) **Unitary regime** ($\sigma \to \infty$ faster than $1/\tau$): the measurement term vanishes; the system evolves freely.
(3) **Lindblad regime** ($\sigma = 1/(k\tau)$, $k > 0$ fixed): both terms contribute at the same order, yielding a non-trivial differential equation.

Master equation from continuous measurement

In the Lindblad regime with $\sigma = 1/(k\tau)$ and decoherence rate $\Gamma = k/8$, the continuous limit of the non-selective evolution is

$$\boxed{\frac{\mathrm{d}\rho_S}{\mathrm{d}t} = -i\,[H_0,\, \rho_S] - \Gamma\,\big[\hat{x}_S, [\hat{x}_S,\, \rho_S]\big]}\,. \tag{162}$$

This is a Lindblad equation with $L \propto \hat{x}_S$; equivalently, $-\Gamma[\hat{x}_S, [\hat{x}_S, \rho_S]] = 2\Gamma\,\mathcal{D}[\hat{x}_S]\,\rho_S$.

Balancing information and disturbance

Each meter in the Lindblad regime is very broad ($\sigma \propto 1/\tau \to \infty$) and extracts almost no information. But infinitely many such measurements accumulate a finite back-action, producing steady decoherence—precisely the mechanism by which a thermal environment induces decoherence.

8. Quantum stochastic calculus (discrete case)

8.1. Free particle under continuous position measurement. We illustrate (162) for a free particle with $H_0 = \hat{p}_S^2/(2m)$.

8.1.1. *First moments.* Using the canonical commutation relations $[\hat{x}_S, \hat{p}_S] = i$, the Ehrenfest equations are

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle\hat{x}_S\rangle = \frac{\langle\hat{p}_S\rangle}{m}\,, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\langle\hat{p}_S\rangle = 0\,. \tag{163}$$

These are identical to the unitary case: the first moments are unaffected by the dephasing.

8.1.2. *Momentum diffusion.* The effect of continuous measurement appears in the second moments. Computing $\frac{\mathrm{d}}{\mathrm{d}t}\langle \hat{p}_S^2\rangle = \mathrm{tr}(\hat{p}_S^2\,\dot{\rho}_S)$ yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathrm{Var}(\hat{p}_S) = 2\Gamma \quad . \tag{164}$$

The momentum variance grows linearly: continuous position monitoring induces **momentum diffusion**.

> **Heisenberg uncertainty at work**
>
> Monitoring position makes it more definite but, by the uncertainty principle, makes momentum *less* definite. The diffusion rate 2Γ is set by the measurement strength—the same parameter that controls the decoherence rate.

Exercise 8.1. Verify that $[\hat{x}_S, [\hat{x}_S, \rho_S]]$ in the position representation equals $(x-x')^2\,\rho_S(x,x')$, where $\rho_S(x,x') = \langle x|\rho_S|x'\rangle$. Conclude that diagonal elements are unaffected while off-diagonal elements decay at a rate proportional to $(x-x')^2$.

Exercise 8.2. Starting from (162) with $H_0 = \hat{p}_S^2/(2m)$, derive (163) and (164). *Hint:* use $[\hat{x}_S, \hat{p}_S^n] = i\,n\,\hat{p}_S^{n-1}$ and the cyclic property of the trace.

Exercise 8.3. Derive (160) by evaluating the Gaussian integral in (178) and expanding the result to first order in $1/\sigma$.

8.2. From master equations to superoperators. Every master equation derived so far—radiative damping, cavity decay, quantum dot—has the Lindblad form

$$\frac{\mathrm{d}\rho_S}{\mathrm{d}t} = -i[H_0,\,\rho_S] + \sum_j \gamma_j\Big(L_j\,\rho_S\,L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j,\,\rho_S\}\Big)\,. \tag{165}$$

This is first-order in time and linear in ρ_S, so its solution can be written as the action of a one-parameter family of maps.

Definition 8.1 (Superoperator)**.** A **superoperator** is a linear map $\mathcal{L}\colon M_d(\mathbb{C}) \to M_d(\mathbb{C})$. Operators act on vectors; superoperators act on operators—hence the name.

Denoting the right-hand side of (165) by $\mathcal{L}(\rho_S)$, the master equation becomes $\dot{\rho}_S = \mathcal{L}(\rho_S)$, whose formal solution is

$$\rho_S(t) = e^{t\mathcal{L}}\,\rho_S(0)\,, \tag{166}$$

with the exponential defined by its Taylor series and powers of $\mathcal{L}$ understood as iterated application.

8.2.1. *Vectorisation.* To solve (165) numerically one converts it into a standard matrix equation by *vectorising* the density operator: the $d\times d$ matrix ρ_S is reshaped into a vector of length d^2.

Definition 8.2 (Standard vectorisation)**.** Given $\rho = \sum_{j,k}\rho_{jk}|j\rangle\langle k|$, its vectorisation is

$$|\rho\rangle\!\rangle = \sum_{j,k}\rho_{jk}\,|j\rangle\otimes|k\rangle \;\in\; \mathbb{C}^d\otimes\mathbb{C}^d \quad . \tag{167}$$

Under this map, left and right multiplication become tensor products:

$$A\,\rho \longmapsto (A \otimes \mathbb{I})\,|\rho\rangle\rangle\,, \qquad \rho\,B \longmapsto (\mathbb{I} \otimes B^\top)\,|\rho\rangle\rangle\,. \tag{168}$$

Applying these rules to (165), the master equation becomes $\frac{\mathrm{d}}{\mathrm{d}t}|\rho_S\rangle\rangle = L\,|\rho_S\rangle\rangle$ where L is the $d^2 \times d^2$ matrix

$$L = -i\big(H_0 \otimes \mathbb{I} - \mathbb{I} \otimes H_0^\top\big) + \sum_j \gamma_j \Big(L_j \otimes \bar{L}_j - \tfrac{1}{2} L_j^\dagger L_j \otimes \mathbb{I} - \tfrac{1}{2}\mathbb{I} \otimes L_j^\top \bar{L}_j\Big)\,. \tag{169}$$

Remark. This is how one solves Lindblad equations in practice: exponentiate the $d^2 \times d^2$ matrix L and reshape the result back into a $d \times d$ density operator.

8.3. Dynamical semigroups. We now abstract away the quantum-mechanical context and study families of linear maps indexed by non-negative time that compose multiplicatively.

Definition 8.3 (Dynamical semigroup)**.** A **dynamical semigroup** is a family of linear maps $\{\mathcal{E}_s\colon \mathbb{C}^d \to \mathbb{C}^d\}_{s\geq 0}$ satisfying

(i) **Semigroup property:** $\mathcal{E}_t \circ \mathcal{E}_s = \mathcal{E}_{s+t}$ for all $s, t \geq 0$,
(ii) **Identity at zero:** $\mathcal{E}_0 = \mathrm{id}$.

Remark. Unlike a group, a semigroup does not require invertibility: the inverse $\mathcal{E}_{-s}$ need not exist. Irreversible dynamics naturally form semigroups rather than groups.

Proposition 8.1 (Generator of a continuous semigroup)**.** *If $s \mapsto \mathcal{E}_s$ is continuous, then $\mathcal{E}_s$ is differentiable in s and there exists a matrix L such that*

$$\mathcal{E}_s = e^{sL}\,. \tag{170}$$

*The matrix L is called the **generator** of the semigroup.*

Continuity implies differentiability

For an arbitrary function, continuity does not imply differentiability. The semigroup property changes this: composing a function with itself smooths out irregularities, much as convolution produces smoother results. The constraint $\mathcal{E}_{s+t} = \mathcal{E}_t \circ \mathcal{E}_s$ is strong enough to force the exponential form.

Proof sketch. Define $M(\epsilon) = \int_0^\epsilon \mathcal{E}_s\,\mathrm{d}s$. For small $\epsilon > 0$, continuity and proximity to the identity ensure $M(\epsilon)$ is invertible. Writing $\mathcal{E}_s = \mathcal{E}_s \cdot M(\epsilon) \cdot M(\epsilon)^{-1}$ and using the semigroup property to shift the integration variable, one obtains a representation that is differentiable by the fundamental theorem of calculus. The derivative $\mathcal{E}'(0)$ is independent of s by the semigroup property, giving $\mathcal{E}_s = e^{sL}$ with $L = \mathcal{E}'(0)$. □

8.4. Quantum dynamical semigroups.

Definition 8.4 (Quantum dynamical semigroup)**.** A family of superoperators $\{\mathcal{E}_s\colon M_d(\mathbb{C}) \to M_d(\mathbb{C})\}_{s\geq 0}$ is a **completely positive quantum dynamical semigroup** (CP semigroup) if

(i) $\mathcal{E}_0 = \mathrm{id}$,
(ii) $\mathcal{E}_t \circ \mathcal{E}_s = \mathcal{E}_{s+t}$ for all $s, t \geq 0$,
(iii) $\mathcal{E}_s$ is completely positive and trace-preserving for every $s \geq 0$.

By vectorisation, any QDS is a dynamical semigroup on $\mathbb{C}^{d^2}$ and therefore admits a generator $\mathcal{L}$ with $\mathcal{E}_s = e^{s\mathcal{L}}$. The central question is: *which generators $\mathcal{L}$ produce completely positive semigroups?*

Why complete positivity is surprising

In deriving master equations we performed Taylor expansions and truncations—operations that generically destroy positivity. A truncated pure state $|\tilde{\psi}(t)\rangle \approx (1+c_0 t)|\psi\rangle$ is typically not normalised, let alone physical. That the resulting evolution is nonetheless CP for all $t \geq 0$ is a non-trivial structural property of the Lindblad form.

8.5. Characterising CP generators.

Definition 8.5 (Conditionally completely positive map)**.** A hermiticity-preserving superoperator $\mathcal{L}$ is **conditionally completely positive** (CCP) if there exists a CP map $\Phi\colon M_d(\mathbb{C}) \to M_d(\mathbb{C})$ and $K \in M_d(\mathbb{C})$ such that

$$\mathcal{L}(\rho) = \Phi(\rho) - K\rho - \rho K^\dagger . \tag{171}$$

Proposition 8.2 (Equivalent characterisation of CCP maps)**.** *Let $|\Omega\rangle = \frac{1}{\sqrt{d}}\sum_j |j\rangle \otimes |j\rangle$ be maximally entangled and $P = \mathbb{I} - |\Omega\rangle\langle\Omega|$. A hermiticity-preserving $\mathcal{L}$ is CCP if and only if*

$$P\left[(\mathcal{L}\otimes \mathrm{id})\,|\Omega\rangle\langle\Omega|\right] P \;\geq\; 0 . \tag{172}$$

Proof sketch. ($\Rightarrow$) Substitute the CCP form into the left-hand side: the Φ term is positive by complete positivity, while the K terms vanish because $P|\Omega\rangle = 0$.

($\Leftarrow$) The Choi operator $\tau = (\mathcal{L}\otimes \mathrm{id})|\Omega\rangle\langle\Omega|$ has a positive block $P\tau P \geq 0$. By the Choi–Jamiołkowski isomorphism this block corresponds to a CP map Φ; the remaining terms take the form $-K\rho - \rho K^\dagger$. □

8.6. The GKSL theorem.

Proposition 8.3 (CP semigroup $\Leftrightarrow$ CCP generator)**.** *Let $\{\mathcal{E}_s\}_{s\geq 0}$ be a continuous semigroup of superoperators with generator $\mathcal{L}$. Then $\mathcal{E}_s$ is completely positive for all $s \geq 0$ if and only if $\mathcal{L}$ is conditionally completely positive.*

Proof sketch. ($\Rightarrow$) Complete positivity of $\mathcal{E}_s$ implies $(\mathcal{E}_s \otimes \mathrm{id})|\Omega\rangle\langle\Omega| \geq 0$. Taylor expand $\mathcal{E}_s = \mathrm{id} + s\mathcal{L} + O(s^2)$ and project with P. Since $P|\Omega\rangle\langle\Omega|P = 0$, the leading term vanishes and at first order in s one obtains the CCP condition (172).

($\Leftarrow$) Decompose $\mathcal{L} = \Phi - \mathcal{K}$ with Φ completely positive and $\mathcal{K}(\rho) = K\rho + \rho K^\dagger$. By the Lie–Trotter formula, $e^{s\mathcal{L}} = \lim_{n\to\infty}(e^{s\Phi/n}\, e^{-s\mathcal{K}/n})^n$. Each factor is CP: $e^{s\Phi/n}$ because the exponential of a CP map is CP, and $e^{-s\mathcal{K}/n}(\rho) = M\rho M^\dagger$ with $M = e^{-sK/n}$. Since compositions and limits of CP maps are CP, the semigroup is CP. □

Complete positivity from the generator

Verifying that a continuous semigroup of superoperators is completely positive reduces to a *single linear-algebraic condition* on the generator: the CCP property (172). This is remarkable because complete positivity of $\mathcal{E}_s$ for every $s > 0$ is, a priori, an infinite family of constraints.

Combining Proposition 8.3 with Proposition 8.2 and imposing trace preservation yields the celebrated result of Gorini–Kossakowski–Sudarshan and Lindblad.

Theorem 8.4 (GKSL theorem — finite-dimensional case)**.** *A superoperator $\mathcal{L}\colon M_d(\mathbb{C}) \to M_d(\mathbb{C})$ generates a continuous completely positive trace-preserving semigroup if and only if*

it has the ***Lindblad form***

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_{j=1}^{d^2-1} \gamma_j \Big(L_j \, \rho \, L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \Big) \quad , \tag{173}$$

where $H = H^\dagger$, the rates satisfy $\gamma_j \geq 0$, and $\{L_j\}$ are operators on $\mathbb{C}^d$ (at most $d^2 - 1$ of them).

Gorini–Kossakowski–Sudarshan and Lindblad (1976)

The finite-dimensional result was obtained independently by Gorini, Kossakowski, and Sudarshan [**?**] and by Lindblad [**?**]. Lindblad proved the more general result for norm-continuous semigroups on $\mathcal{B}(\mathcal{H})$ in arbitrary dimensions.

Exercise 8.4. Starting from the Lindblad equation for a single jump operator L with rate γ and Hamiltonian H, derive the explicit $d^2 \times d^2$ matrix in (169) using the vectorisation rules (168).

Exercise 8.5. Let $\Phi\colon M_d(\mathbb{C}) \to M_d(\mathbb{C})$ be completely positive. Show that $e^{s\Phi} = \sum_{n=0}^{\infty} \frac{s^n}{n!} \Phi^n$ is completely positive for all $s \geq 0$.
Hint: The composition of CP maps is CP, and a convergent sum of positive operators is positive.

Exercise 8.6. Verify the Lie–Trotter formula $e^{t(A+B)} = \lim_{n\to\infty} (e^{tA/n}\, e^{tB/n})^n$ by Taylor expanding both sides to first order in t/n and bounding the error.

8.7. The GKSL theorem. We have seen that every master equation derived so far has the Lindblad form (146). Last lecture we showed that if $\{\mathcal{E}_s\}_{s\geq 0}$ is a dynamical semigroup ($\mathcal{E}_0 = \mathrm{id}$, $\mathcal{E}_s \circ \mathcal{E}_t = \mathcal{E}_{s+t}$) that is linear and completely positive, then $\mathcal{E}_s = e^{s\mathcal{L}}$ with $\mathcal{L}(\rho) = \Phi(\rho) - K\rho - \rho K^\dagger$ for a CP map Φ and operator K. It remains to impose *trace preservation*.

Theorem 8.5 (Gorini–Kossakowski–Sudarshan–Lindblad)**.** *Let $\{\mathcal{E}_s\}_{s\geq 0}$ be a continuous semigroup of CPTP maps on a finite-dimensional system. Then $\mathcal{E}_s = e^{s\mathcal{L}}$ with*

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_j \Big(L_j \, \rho \, L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \Big) \quad , \tag{174}$$

where $H = H^\dagger$ and $\{L_j\}$ are the Lindblad operators.

Proof sketch. Trace preservation requires $\mathrm{tr}\big(\mathcal{L}(\rho)\big) = 0$ for all ρ. Using $\mathcal{L}(\rho) = \Phi(\rho) - K\rho - \rho K^\dagger$ and the adjoint map Φ^* (defined by $\mathrm{tr}(\Phi^*(X)\,\rho) = \mathrm{tr}(X\,\Phi(\rho))$), this becomes

$$\Phi^*(\mathbb{I}) = K + K^\dagger \, . \tag{175}$$

The Kraus theorem gives $\Phi(\rho) = \sum_j L_j \, \rho \, L_j^\dagger$, so $\Phi^*(\mathbb{I}) = \sum_j L_j^\dagger L_j$. Hence $K + K^\dagger = \sum_j L_j^\dagger L_j$. Decomposing K into Hermitian and anti-Hermitian parts—just as a complex number splits into real and imaginary parts—we set $H = -\frac{i}{2}(K - K^\dagger)$ and obtain $K = \frac{1}{2}\sum_j L_j^\dagger L_j + iH$. Substituting back yields (174). □

Physical significance

The theorem guarantees that any equation of motion with Lindblad structure is a physically implementable process: it maps density operators to density operators for all $t \geq 0$. This

is non-trivial—the Born–Markov approximation truncates a series, and there is no *a priori* reason for the truncated equation to preserve complete positivity.

The inverse question

So far we have derived Lindblad equations from unitary system–environment models. Can we go the other way? Given an arbitrary Lindblad equation, does there exist a unitary dilation that reproduces it? The answer is *yes*, and quantum jumps provide an explicit construction.

9. Quantum trajectories and quantum jumps

9.1. Dichotomic measurement model. Consider the single-jump-operator Lindblad equation

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -i[H, \rho] + C\,\rho\,C^\dagger - \tfrac{1}{2}\{C^\dagger C, \rho\}\,, \tag{176}$$

with C a single jump operator (generalisation to multiple L_j is straightforward). We model a detector that continuously interrogates the system in time bins of width $\mathrm{d}t$, reporting either "click" or "no click".

Definition 9.1 (Measurement operators)**.** The dichotomic instrument is defined by

$$M_0 = \mathbb{I} - \left(iH + \tfrac{1}{2}C^\dagger C\right)\mathrm{d}t \quad , \qquad M_1 = C\,\sqrt{\mathrm{d}t}\,. \tag{177}$$

These satisfy $M_0^\dagger M_0 + M_1^\dagger M_1 = \mathbb{I} + \mathcal{O}(\mathrm{d}t^2)$, forming a valid POVM to first order.

Remark. Setting $\hat{R} = C^\dagger C$, the operator $\hat{R}$ is the number operator for the relevant excitations (e.g. photons in an output beam).

9.2. Non-selective evolution. When the measurement outcome is *not* recorded, the state updates as

$$\rho(t + \mathrm{d}t) = M_0\,\rho\,M_0^\dagger + M_1\,\rho\,M_1^\dagger\,. \tag{178}$$

Expanding to order $\mathrm{d}t$ recovers exactly the Lindblad equation (176), confirming that this measurement model is a valid dilation.

9.3. Selective evolution. When we *look* at the measurement record, the detection probabilities in a window $\mathrm{d}t$ are

$$p_1 = \mathrm{tr}(C^\dagger C\,\rho)\,\mathrm{d}t\,, \qquad p_0 = 1 - p_1\,. \tag{179}$$

Poisson statistics

The click probability is proportional to $\mathrm{d}t$: as $\mathrm{d}t \to 0$ the per-bin probability vanishes, but the number of bins in a finite interval grows, so a finite average number of clicks occurs—the hallmark of a **Poisson process**.

If the system is in a pure state $|\psi(t)\rangle$ before the measurement, it remains pure afterwards:

State update rules

No click ($\mathrm{d}N = 0$):

$$|\psi_0(t+\mathrm{d}t)\rangle = |\psi(t)\rangle - \Big(iH + \tfrac{1}{2}C^\dagger C - \tfrac{1}{2}\langle C^\dagger C\rangle_t\Big)|\psi(t)\rangle\,\mathrm{d}t + \mathcal{O}(\mathrm{d}t^2)\,, \tag{180}$$

where $\langle C^\dagger C\rangle_t = \langle\psi(t)|C^\dagger C|\psi(t)\rangle$.
Click ($\mathrm{d}N = 1$)—the **quantum jump**:

$$|\psi_1(t+\mathrm{d}t)\rangle = \frac{C|\psi(t)\rangle}{\|C|\psi(t)\rangle\|}\,. \tag{181}$$

Information from silence

Even when the detector does *not* click, information is gained: the absence of a click updates the state via the $-\frac{1}{2}C^\dagger C + \frac{1}{2}\langle C^\dagger C\rangle_t$ correction. Seeing nothing is not the same as learning nothing.

9.4. Stochastic Schrödinger equation. Let $N(t)$ count the total clicks in $[0,t]$. The increment $\mathrm{d}N(t) \in \{0,1\}$ satisfies

$$\mathrm{d}N(t)^2 = \mathrm{d}N(t)\,, \qquad \mathbb{E}[\mathrm{d}N(t)] = \langle C^\dagger C\rangle_t\,\mathrm{d}t\,. \tag{182}$$

Combining both branches and using $\mathrm{d}N\,\mathrm{d}t = \mathcal{O}(\mathrm{d}t^2)$, the conditional pure-state evolution is

$$\mathrm{d}|\psi\rangle = -\Big(iH + \tfrac{1}{2}C^\dagger C - \tfrac{1}{2}\langle C^\dagger C\rangle_t\Big)|\psi\rangle\,\mathrm{d}t + \left(\frac{C}{\sqrt{\langle C^\dagger C\rangle_t}} - \mathbb{I}\right)|\psi\rangle\,\mathrm{d}N(t)\,. \tag{183}$$

The first term is deterministic drift; the second activates at each click and implements the quantum jump.

9.5. Quantum jump trajectories. A realisation of $N(t)$—a staircase with jumps at random times $t_1 < t_2 < \cdots$—defines a **quantum jump trajectory**. Between clicks the state evolves under the non-Hermitian effective Hamiltonian $H_{\mathrm{eff}} = H - \frac{i}{2}C^\dagger C$; at each click the operator C is applied:

$$|\psi(t)\rangle \propto \mathcal{T}\exp\left(-i\int_{t_k}^{t} H_{\mathrm{eff}}\,\mathrm{d}s\right) C\;\mathcal{T}\exp\left(-i\int_{t_{k-1}}^{t_k} H_{\mathrm{eff}}\,\mathrm{d}s\right) C\;\cdots\;|\psi(0)\rangle\,, \tag{184}$$

where $\mathcal{T}$ denotes time ordering.

Recovering the Lindblad equation

Averaging over all measurement records recovers the Lindblad dynamics:

$$\rho(t) = \mathbb{E}\big[|\psi(t)\rangle\langle\psi(t)|\big] = e^{t\mathcal{L}}\rho(0)\,. \tag{185}$$

The non-selective evolution is the statistical mixture of all selective quantum jump trajectories.

Exercise 9.1. Verify (178) reproduces (176) by expanding to order $\mathrm{d}t$.

Exercise 9.2. From the stochastic Schrödinger equation (183), compute $\mathrm{d}|\psi\rangle\langle\psi|$ to order $\mathrm{d}t$ using $\mathrm{d}N^2 = \mathrm{d}N$ and $\mathrm{d}N\,\mathrm{d}t \approx 0$. Take the expectation and recover the Lindblad equation for $\rho = \mathbb{E}[|\psi\rangle\langle\psi|]$.

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